

Integrative analysis of DNA replication origins and ORC/MCM binding sites in human cells reveals a lack of overlap

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Abstract

Based on experimentally determined average inter-origin distances of ~100 kb, DNA replication initiates from ~50,000 origins on human chromosomes in each cell cycle. The origins are believed to be specified by binding of factors like the Origin Recognition Complex (ORC) or CTCF or other features like G-quadruplexes. We have performed an integrative analysis of 113 genome-wide human origin profiles (from five different techniques) and 5 ORC-binding profiles to critically evaluate whether the most reproducible origins are specified by these features. Out of ~7.5 million union origins identified by all datasets, only 0.27% were reproducibly obtained in at least 20 independent SNS-seq datasets and contained in initiation zones identified by each of three other techniques (20,250 shared origins), suggesting extensive variability in origin usage and identification. 21% of the shared origins overlap with transcriptional promoters, posing a conundrum. Although the shared origins overlap more than union origins with constitutive CTCF binding sites, G-quadruplex sites and activating histone marks, these overlaps are comparable or less than that of known Transcription Start Sites, so that these features could be enriched in origins because of the overlap of origins with epigenetically open, promoter-like sequences. Only 6.4% of the 20,250 shared origins were within 1 kb from any of the ~13,000 reproducible ORC binding sites in human cancer cells, and only 4.5% were within 1 kb of the ~11,000 union MCM2-7 binding sites in contrast to the nearly 100% overlap in the two comparisons in the yeast, *S. cerevisiae*. Thus, in human cancer cell lines, replication origins appear to be specified by highly variable stochastic events dependent on the high epigenetic accessibility around promoters, without extensive overlap between the most reproducible origins and currently known ORC- or MCM-binding sites.

eLife assessment

The paper contains some **useful** analysis of existing data but there are concerns regarding the conclusion that there might be alternative mechanisms for determining the location of origins of DNA replication in human cells compared to the well known mechanism known from many eukaryotic systems, including yeast, *Xenopus*, *C. elegans* and *Drosophila*. The lack of overlap between binding sites for ORC1 and ORC2, which are known to form a complex in human cells, is a particular concern and points to the evidence for the accurate localization of their binding sites in the genome being **incomplete**.

Introduction

DNA replication is essential for the duplication of a cell and the maintenance of the eukaryotic genome. To replicate the human diploid genome of ~3 billion base pairs, efficient cellular programs are coordinated to ensure genetic information is accurately copied. In each human cell cycle, replication starts from ~50,000 genomic locations called replication origins (Hu and Stillman 2023 [↗](#)). At an origin of replication, the double-stranded DNA is unwound, and primase-DNA polymerase alpha lays down the first RNA primer extended as DNA. Once replication is initiated, the helicase involved in unwinding the origin continues to unwind DNA on either side and the replication proteins, including DNA polymerases, help copy the single-stranded DNA. Unlike yeast origins, human and most eukaryotic origins do not seem to have a clear DNA sequence preference (Hu and Stillman 2023 [↗](#)), (Leonard and Méchali 2013 [↗](#)). Yet it becomes clear that chromatin context and other DNA-based activity are important factors for human and non-yeast eukaryotic origin selection. A great amount of effort has been spent to understand how human origins are specified and sometimes different trends are observed regarding genomic features enriched at human origins. For example, G-quadruplex motifs were found to be associated with human origins (Besnard et al. 2012 [↗](#)), but can only explain a small subset of origins even after controlling for technical biases (Foulk et al. 2015 [↗](#)). To better resolve these issues, we reasoned that a systematic analysis of all available human origin-mapping datasets will help uncover potential technical and biological details that affect origin detection.

To profile replication origins genome-wide, several sequencing-based methods have been developed, including Short Nascent Strand (SNS)-seq (Foulk et al. 2015 [↗](#)), Repli-seq (Hansen et al. 2010 [↗](#)), Rerep-seq (Menzel et al. 2020 [↗](#)), and Bubble-seq (Mesner et al. 2013 [↗](#)). SNS-seq assays across different human cell types identified a subset of origins that can explain ~80% of origins in any tested cell type (Akerman et al. 2020 [↗](#)), however no study has systematically examined the consistency within and between different origin-mapping techniques. These methods are based on different molecular capture strategies and therefore are expected to have technical biases. For instance, SNS-seq utilizes lambda exonuclease (λ -exo) to enrich for RNA-primed newly replicated DNA by removing parental DNA fragments. However, the products may be contaminated with chromosomal DNA fragments or be affected by the cutting bias of λ -exo against GC-rich sequences (Foulk et al. 2015 [↗](#)), although it has been proposed that one can experimentally correct and control for such biases (Akerman et al. 2020 [↗](#)). SNS-seq is the only method that yields origins at a resolution of a few hundred base-pairs, with all other methods delineating larger initiation zones (IZ) that may have multiple origins. Repli-seq relies on nucleotide pulse-labeling and antibody enrichment of newly replicated DNA superimposed on cells fractionated at different parts of S phase by flow cytometry. Here the results may be contaminated by DNA non-specifically associated with the antibody (Zhao et al. 2020 [↗](#)). Rerep-seq only captures the new synthesized sequences that replicate more than once when replication is dysregulated, and these origins tend

to be enriched in the early replicating, epigenetically open parts of the genome (Menzel et al. 2020). Bubble-seq (Mesner et al. 2013) generates long reads because it captures the origin-containing replication intermediates by selection of DNA bubbles, and so may enrich for origins that are flanked by pause sites. Okazaki-seq (OK-seq) identifies sites where the direction of the Okazaki fragments changes from left-ward to right-ward on the chromosome revealing sites where the two lagging strands diverge from each other (Petryk et al. 2016). Because the technical biases mentioned above are associated with specific techniques, we suspect that origins captured across different techniques are less likely to be affected by those biases. Given the large number of datasets in the public domain (>100 datasets for human origins), this is an opportune time to study the reproducibility between the studies and determine the most reproducible and consistent origins identified by SNS-seq and confirmed by each of the other methods. This most reproducible group of origins (shared origins) from different research groups using different techniques and different cell lines, will be far fewer than all the origins reported to date and minimize complications from stochastic (noisy) firing of origins, but they are best expected to fit the current hypotheses regarding origin specification.

Yeast origin recognition complex (ORC) binds to double-stranded DNA with sequence specificity, helps to load minichromosome maintenance protein complex (MCM) and thus prepare the origins for subsequent firing (Bell and Dutta 2002; Remus et al. 2009; Bell and Stillman 1992), (Costa and Diffley 2022). Consistent with this, a role of ORC in loading MCM proteins was also described in *Xenopus* egg extracts (Rowles et al. 1999; Coleman et al. 1996). All of this leads to the expectation that in human cells there will be significant concordance between ORC binding sites and efficient origins of replication. Efforts to define double-stranded DNA sequences bound specifically by human ORC find very little sequence specificity (Vashee et al. 2003; Hoshina et al. 2013), and this may be responsible for the lack of sequence specificity in human origins. This difference between yeast and human ORC is attributed to sequence features in ORC4, one of the subunits of the origin recognition complex (Lee et al. 2021). When the yeast ORC was humanized by appropriate mutation in the ORC4 subunit, the sequence specificity of ORC binding was lost even in yeast. Despite this loss of sequence specificity, the mutant yeast ORC loaded MCM proteins, and the yeast replicated DNA and survived. Thus, even if human origins of replication do not have specific sequences, one should expect some concordance between ORC binding sites and the most reproducible origins of replication. An additional complexity is that genome-wide analysis found active origins and dormant origins determined by SNS-seq and OK-seq have little difference in ORC or MCM density (Sugimoto et al. 2018; Kirstein et al. 2021). Finally, a few publications have reported significant replication after genetic mutation of *Drosophila* and human ORC subunit genes that reduce the expressed proteins to near undetectable levels (Park and Asano 2008; Shibata et al. 2016; Okano-Uchida et al. 2018; Shibata and Dutta 2020). Definition of the most reproducible and consistent origins identified by different methods from different research groups in different cell lines thus provides a unique opportunity to determine how much overlap is seen between the highly reproducible human origins and the ORC-binding sites reported in the literature from ChIP-seq studies.

To understand the genome-wide distribution patterns of replication origins in an unbiased way, we performed an integrative analysis of 113 DNA replication origin datasets. Because SNS-seq is unique in identifying origins of a few hundred bases, while the other methods identify larger initiation zones that contain origins, we first prepared a list of high-resolution SNS-seq origins that have been identified in at least 20 of the 66 SNS-seq datasets in this study. We next determined those reproducible SNS-seq origins that overlap with initiation zones identified by each of the three techniques (Repli-seq, OK-seq and Bubble-seq) to identify 20,250 high-confidence origins that are shared between SNS-seq and every other method identifying initiation zones. Using these shared origins, which overlap significantly with transcription promoters, we tested whether G-quadruplex sites, CTCF binding sites, ORC binding sites or MCM binding sites help specify origins.

A total of 7,459,709 origins from 113 datasets show similar but different genomic features associated with each origin-mapping technique

We collected 113 publicly available replication origin identification studies in different human cell types from five different techniques (Supplementary Figure S1a), including SNS-seq, Repli-seq, Rerep-seq, OK-seq and Bubble-seq. The complete list of datasets used in this analysis can be found in Supplementary Table 1. We processed all the datasets using the first two steps of a pipeline with different parameters considering the various resolution of different techniques, as SNS-seq is unique in identifying high-resolution origins of replication (**Fig. 1a**). Each of the 113 datasets yielded at least 1000 origins. We merged origins that overlap for at least 1bp from each other and cut the merged regions into 300bp segments, considering origin lengths were significantly longer for Bubble-seq and OK-seq, methods known to identify initiation zones (Supplementary Fig. S1b). A total of ~7,460,000 union origins were discovered from all techniques (**Fig. 1b**).

Principal Component Analysis (PCA) of all origin datasets shows that origin locations from the same technique are more similar to each other than from different techniques (**Fig. 1c**). This is confirmed by pair-wise correlations of the datasets, where it is evident that each method identifies origins that are best correlated with origins identified by that method alone (Supplementary Fig. S3). Although the most popular technique, SNS-seq, shows some variability in the PCA (**Fig. 1c**), it is also the one that is best correlated with origins identified by other SNS-seq datasets. Re-rep-seq and Bubble-seq seemed to identify initiation zones that were also better correlated with SNS-seq origins (Supplementary Fig. S3). Since Re-rep-seq tends to identify origins that are enriched in early replicating, gene-rich parts of the genome, this result suggests that SNS-seq and Bubble-seq also have some bias for origins in early replicating, gene-rich parts of the genome. However, we did not observe clear trends of clustering of origins identified in similar cell types (Supplementary Fig. S1c), even when we focused only on SNS-seq origins (Supplementary Fig. S1d). The one exception was the SNS-seq origins from T lymphoblasts, which were very closely clustered with each other in six datasets, but these were all done in a the CCRF-CEM cell-line (or a derivative) by one group at the same time (Murai et al. 2018). We only examined SNS-seq data from 2018 onwards, when the lambda exonuclease-based digestion step had been incorporated into SNS-seq protocols, but wondered if there was a steady improvement of reproducibility as one progressed through the years. However, SNS-seq datasets do not become progressively more reproducible as one goes year-by-year from 2018-2022 (Supplementary Fig. S1e). The results suggest that there is significant difference in origins captured by different techniques that cannot be explained by differences in cell choices. For the most popular technique (SNS-seq) the variability may be least when confined to a single cell line studied at the same time by the same group.

We next checked the genomic characteristics of origins defined by each technique. Regardless of technique used, around half of the detected origins fall in intergenic regions, followed by 30-40% allocated to introns (**Fig. 1d**). We recently reported 40,110 genomic regions are routinely bound by more than 3000 TF ChIP-seq data, named as TF binding hotspots (Hu et al. 2021). These TF binding hotspots likely represent highly active open chromatin regions. Despite the differences among techniques, the origins in general have lower overlap with TF binding hotspot (<0.5%), compared to the 16% overlap of gene promoter regions (TSS +/-100bp) with such TF binding hotspots (**Fig. 1e**). We further focused on one specific TF, CCCTC-binding factor (CTCF), which is important in organizing DNA structure and reported to be associated with replication (Su et al. 2020). Origins show significantly lower overlap with constitutive CTCF sites, defined as those that are conserved across cell types (Fang et al. 2020), compared to promoters (**Fig. 1f**). G-quadruplexes are found to be correlated with both replication and transcription (Lipps and Rhodes 2009). However, compared to promoters, origins defined by these techniques show

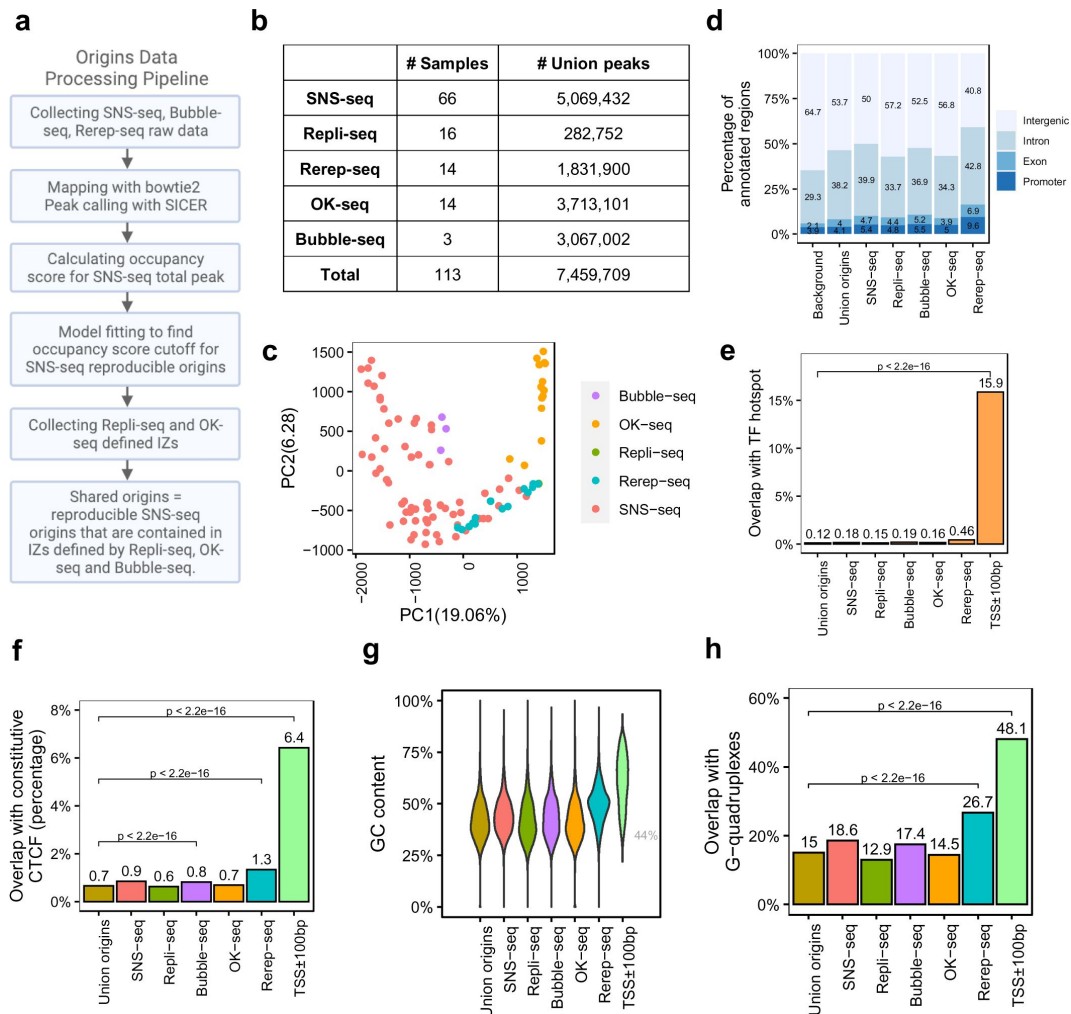


Figure 1

Total 7,459,709 origins defined by four types of techniques show different genomic features.

(a) Data processing pipeline. 113 publicly available profiles of origins are processed following the pipeline. **(b)** Number of samples collected for each technique. In total 7,459,709 union origins were identified. **(c)** PCA shows the clustering of origin datasets from different techniques. **(d)** Genomic annotation (TSS, exon, intron and intergenic regions) of different groups of origins. Background is the percentage of each annotation on whole genome. **(e)** Overlap with TF hotspots for different groups of origins. **(f)** Overlap with constitutive CTCF binding sites for different groups of origins. **(g)** GC content of different groups of origins. Grey line marks the average GC content of human genome. **(h)** G-quadruplex overlapping rates of different group of origins.

significantly lower GC content (**Fig. 1g**). In addition, there is significantly lower overlap of these origins with G-quadruplex sites identified from predicted G-quadruplex motif regions (Bedrat et al. 2016), compared to the overlap of promoters with G-quadruplex sites (**Fig. 1h**). Thus the reported correlations between CTCF binding sites or G-quadruplexes with origins are not as striking as that of these features with promoters.

We found that all origins, regardless of technique by which identified, are slightly enriched at gene promoters, exons, or introns, compared with the genome background (**Fig. 1d**). Among the five different techniques, origins defined by Rerep-seq show the highest level of enrichment with promoters, TF hotspots, constitutive CTCF binding sites and G-quadruplexes and the highest GC content. This is consistent with our knowledge that areas of the genome that are re-replicated when the cell-cycle is disturbed, are enriched in parts of the genome that replicate early in S phase, regions which are enriched in transcriptionally active (and thus epigenetically open) genes and their promoters.

Shared origins are associated with active chromatin and transcription regulatory elements

To address the potential biases caused by each technique, we investigated how many of the ~5 million union SNS-seq origins are reproducible in SNS-seq data and confirmed by other sequencing-based techniques. As we will describe below, these shared origins show significant overlap with promoters. Because the Re-rep-seq origins appear to be slightly different from the origins identified by other techniques, with highest overlap with transcriptionally active genes and promoters, we decided to exclude the Re-rep-seq from the analysis of shared origins and still reached the conclusion summarized above.

SNS-seq origins have the highest resolution. We used the following strategy to determine how many independent confirmations of an SNS-seq origin is sufficient for selecting an SNS-seq origin as a reproducible origin. The occupancy score of each origin defined by SNS-seq (Supplementary Fig. S2a) counts the frequency at which a given origin is detected in the datasets under consideration. Plotting the number of union SNS-seq origins with various occupancy scores with all SNS-seq datasets published after 2018, we sought to determine whether the curve deviates from the random background at a given occupancy score (**Fig. 2a**). For the random background, we assumed that the number of origins confirmed by increasing occupancy scores decreases exponentially (see Methods and Supplementary Table 2). The threshold occupancy score, to determine whether an origin is a reproducible origin, is the point where the observed number of origins deviates from the expected background number (with an empirical FDR < 0.1) (**Fig. 2a**), an occupancy score of 20. Thus the reproducible SNS-seq origins (with a FDR<0.1) were those observed in at least 20 SNS-seq datasets.

We next determined which reproducible SNS-seq origins are confirmed by origins from Repli-seq and replication initiation zones (IZs) from Bubble-seq and OK-seq (**Fig. 2b**). 20,250 of the reproducible SNS-seq origins were found to overlap with an origin or IZ identified by each of the three other techniques and were called shared origins. These high-confidence shared origins consist of 0.27% of all ~7.5 million union origins. The coordinates of the shared origins are available in the supplementary files.

The shared origins have a greater overlap rate with gene promoters and exons compared to union origins (**Fig. 2c**). This is in line with the previous observation that replication and transcription are highly coordinated and enrichment of origins at TSS (Ganier et al. 2019; Cook 1999; Karnani et al. 2010). Shared origins also have a substantially higher overlap rate than union origins with TF binding hotspots (**Fig. 2d**). Moreover, shared origins have a higher overlap rate with constitutive CTCF binding sites, compared to union origins (**Fig. 2e**), and have a higher GC content and overlap with G-quadruplexes than union origins (**Fig. 2f-g**).

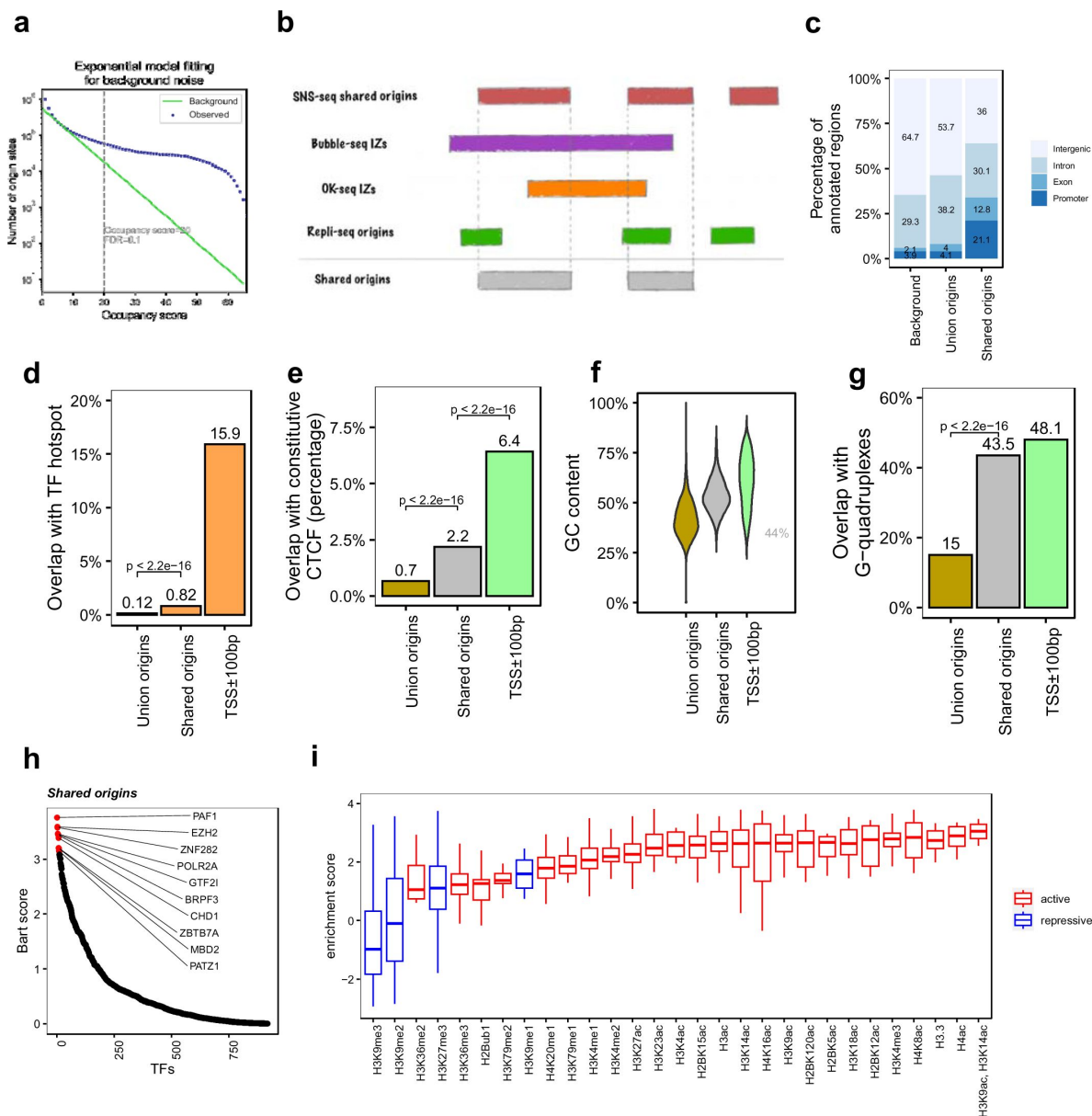


Figure 2

The shared origins are enriched with certain transcription factors and active histone marks.

(a) SNS-seq origins fitting distribution to an exponential model. (b) Conceptual model of how shared origins is calculated. Every SNS-seq shared origin overlaps with any Bubble-seq IZ, OK-seq IZ and Repli-seq origin together is considered as origin shared by all techniques(shared origins). (c) Genomic annotation of union origins and shared origins. (d) Overlap with TF hotspots of union origins and shared origins. (e) Overlap with constitutive CTCF binding sites of union origins and shared origins. (f) GC content of union origins and shared origins. (g) G-quadruplex overlapping rates of union origins and shared origins. (h) BART prediction of TFs associated with shared origins. (i) Enrichment of histone marks at shared origins using all origins as control.

BART (Wang et al. 2018 [↗](#)) analyzes the enrichment of TF binding sites (determined experimentally by ChIP-seq) with areas of interest in the genome. We used BART to perform TF binding site enrichment analysis on the shared origins and identified potential TFs or components of chromatin remodeling factors (e.g., PAF1, EZH2, ZNF282, POLR2A, GTF2I) whose binding sites are associated with the shared origins (**Fig. 2h** [↗](#)). The high enrichment of activators or repressors of transcription in the factors that have binding sites near shared origins provides more support that the shared origins have properties similar to transcriptional promoters.

To investigate whether the shared origins have a specific chromatin epigenomic signature compared to union origins, we used 5,711 publicly available ChIP-seq datasets for 29 different histone modifications and generated a comprehensive map of histone modification enrichment at shared origins compared to union origins. A substantial enrichment of activating histone marks, including H3K4me3 and H3/H4 acetylation, was observed at shared origins compared to union origins (**Fig. 2i** [↗](#)). The enrichment of H3K14ac is interesting given the enrichment in the BART analysis of binding sites of BRPF3, a protein involved in this specific acetylation and reported to stimulate DNA replication (Feng et al. 2016 [↗](#)), but this modification was not uniquely enriched at the shared origins. These results show that as we move from all origins to a small set of high-confidence shared origins, we see a progressive increase in enrichment of TSS in epigenetically active parts of the genome.

However, transcriptional and epigenetic activators are not the whole story. H3K27me3, a repressive mark, is also enriched at shared origins, although this enrichment is not as high as that of the activating marks. The enrichment of EZH2 binding sites near shared origins is consistent with this observation because EZH2 is part of the Polycomb Repressive Complex 2 known to be a writer of H3K27me3. The paradoxical enrichment of repressive marks is consistent with the shared origins also being near binding sites of MBD (binds methylated DNA and represses transcription), PATZ1 and ZNF282, proteins known to be repressors of transcription.

Having identified a small group of highly reproducible origins (independent of cell-type and technique) we asked whether in a given cell line was one technique superior to the others in identifying such origins. K562 cells have been interrogated for origins by three different techniques: SNS-seq, OK-seq and Repli-seq (Supplementary Fig. S4). Among the 20,250 high-confidence shared origins, 9,901 (48.9%) overlapped with SNS-seq origins in K562; 3,872 (19.1%) overlapped with OK-seq IZs in K562; 1,163 (5.7%) overlapped with Repli-seq origins in K562. This suggests that in one cell line, even the large initiation zones defined by OK-seq or Repli-seq do not capture most of the high-confidence origins. In the opposite direction, where we estimate what fraction of origins found by a given technique fall in a reproducible origin, the opposite result emerges: 2.7% of SNS-seq origins, 17.2% of OK-seq initiation zones and 7.7% of Repli-seq initiation zones overlapped with the 20,250 shared origins (Supplementary Fig. S4). Thus SNS-seq may be able to identify more of the reproducible origins, but it comes with a high false positive rate.

Human, but not yeast, high-confidence origins have low overlap with known ORC binding sites

Origin recognition complex (ORC) is expected to bind near replication origins during the cell cycle to help define origins (Bell 2002 [↗](#)). To investigate the correlation of ORC binding with shared origins detected across different origin-mapping techniques, we analyzed all five publicly available ORC1 and ORC2 ChIP-seq datasets with at least 1,000 peaks (Supplementary Table 1c). We identified a total of 34,894 ORC binding sites in the human genome (Supplementary Table 3). Union (all) of ORC binding sites are enriched at promoters, TF hotspots, constitutive CTCF sites, GC content and G-quadruplexes compared to randomized genome background (**Fig. 3a-e** [↗](#)), somewhat similar to what we observed for shared origins (**Figure 2** [↗](#)).

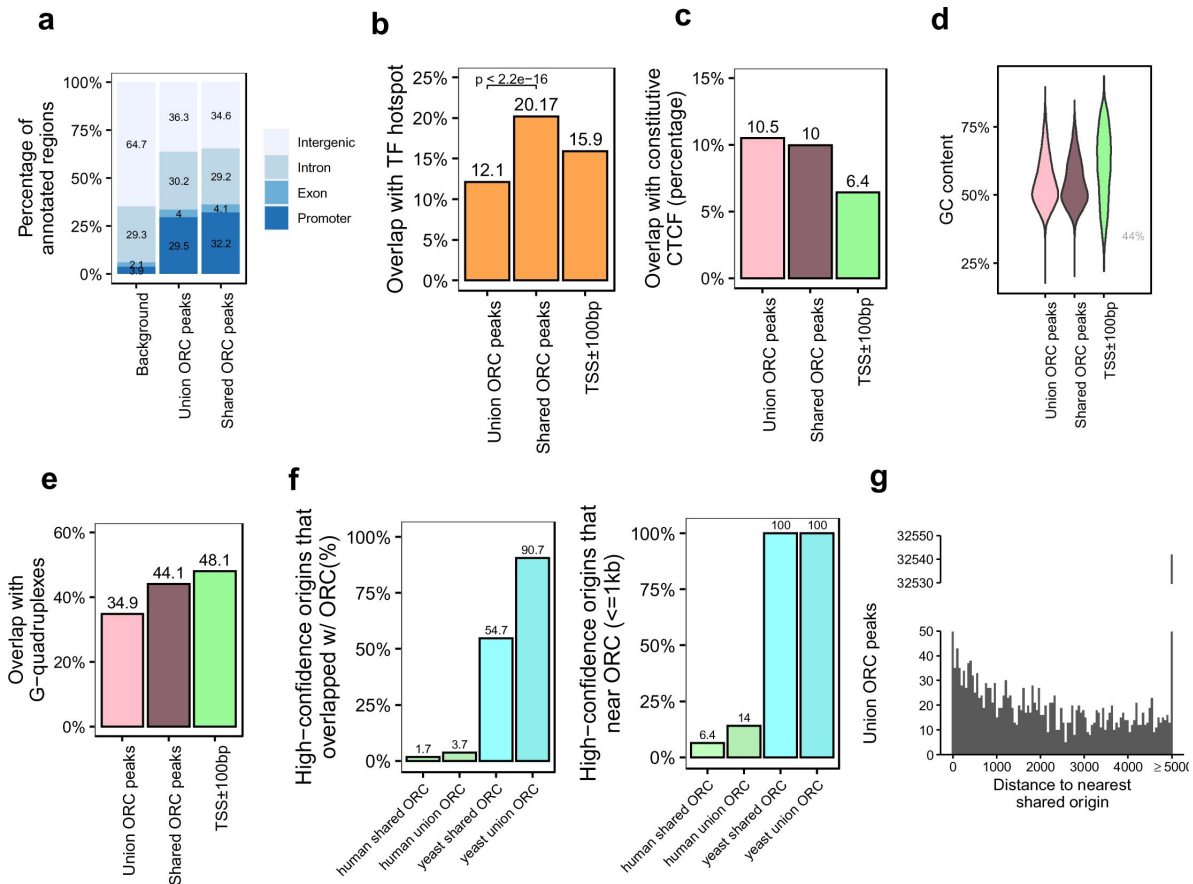


Figure 3

Genomic features of shared ORC binding sites.

(a) Genomic annotation of union ORC and shared ORC binding sites. **(b)** Overlap with TF hotspot of union ORC and shared ORC binding sites. **(c)** Overlap with constitutive CTCF binding rates of union ORC and shared ORC binding sites. **(d)** GC content of union ORC and shared ORC binding sites. **(e)** Overlap with G-quadruplex of union ORC and shared ORC binding sites. **(f)** The percentage of high-confidence origins (shared origins in human and confirmed origins in yeast) that overlapped with or near (within 1 kb) corresponding ORC binding sites. **(g)** Distribution of the distance between ORC binding sites and the nearest shared origin.

Of the 34,894 ORC binding sites in the human genome 12,712 sites were defined as shared ORC binding sites as they occur in at least two ORC datasets (Occupancy score ≥ 2). For the majority of genomic features, including enrichment at promoter (**Fig. 3a**), overlap with constitutive CTCF sites (**Fig. 3c**), GC content (**Fig. 3d**) and overlap with G-quadruplex (**Fig. 3e**), ORC binding sites have similar genomic distribution regardless of how many samples they appear in. Interestingly, the G-quadruplex enrichment at either union or shared ORC binding sites is a bit lower than in gene promoter regions (**Fig. 3e**). The only significant difference between the union ORC sites and the shared ORC binding sites is the significantly higher overlap of the latter with TF binding hotspots (**Fig. 3b**), suggesting that ORC-ChIP seq data also tend to enrich for highly open chromatin regions.

Based on the assumption that ORC binding sites that appeared in more than one dataset are likely to be true-positive ORC binding sites, we analyzed how many of the 12,712 shared ORC binding sites *overlap* with the 20,250 shared origins identified by our analysis above. The overlap was surprisingly low: 1.7% of the shared origins overlapped with shared ORC binding sites (**Fig. 3f**, Supplementary Fig. S5a). Even when we relaxed the criteria to look at shared origins that are proximal (≤ 1 kb) to shared ORC binding sites only 1300 (6.4%) of the shared origins were proximal to the shared ORC binding sites (**Fig. 3f**) (Supplementary Table 4b). The low degree of overlap or proximity of shared origins with shared ORC binding sites indicates that the vast majority of the shared origins are not near experimentally determined ORC binding sites.

In the reverse direction, we asked whether an ORC binding site definitively predicts the presence of an origin nearby. A histogram of the distance between any ORC binding site (union) and the nearest shared origin showed that only 1,086 (3.11%) ORC binding sites are proximal to (≤ 1 kb) a shared replication origin (**Fig. 3g**, Supplementary Table 4a). Given there are 34,894 ORC binding site and 20,250 shared origins, if ORC binding was sufficient to determine a high confidence origin, nearly half of the ORC binding sites should have been proximate to the shared origins. This low level of proximity between ORC binding and reproducible origins suggests that the current data on ORC binding sites is unable to predict the presence of a high confidence, actively fired origin.

We examined whether the co-localization is better if the analysis is done with data exclusively from the same human cell type (Supplementary Fig. S5b and S5d). Only 8.8% of the 105,881 union origins identified by SNS-seq, OK-seq, or Repli-seq in K562 cells overlap with ORC2 binding sites mapped in the same cell line, and only 4.9% of the 68,003 union SNS-seq origins mapped in HeLa cells overlap with ORC1 binding sites in the same cell line. The overlaps improve marginally if we focus on shared origins: 12.8% of 9,605 shared origins in K562 cells and 6.1% of 3,390 shared origins in HeLa cells overlap with ORC2 and ORC1 ChIP-seq sites in the concordant cell lines.

Experiments in the yeast *S. cerevisiae* have demonstrated that ORC binding is critical for defining an origin of replication. Indeed, in contrast to humans, in the more compact genome of the yeast, *S. cerevisiae*, there is higher overlap or proximity (within 1 kb) seen between the Yeast ORC ChIP-seq binding sites (Supplementary Table 1) and the well mapped yeast origins of replication in OriDB (Nieduszynski et al. 2007). In this case, 100% of the shared origins are proximate to (≤ 1 kb) all (union) or shared ORC binding sites in yeast (**Fig. 3f**).

Overall, we found that the shared origins are highly co-localized with ORC binding sites in yeast but not in the human cell lines, suggesting that the current methods of ORC binding site determination are failing to identify functional ORC binding.

Properties of the 6.4% shared origins co-localized reproduced ORC binding sites

Because of the poor co-localization of shared origins with ORC binding sites, we next focused on the 6.4% of shared origins (1,300) that are proximate to a shared ORC binding site, a group we will call the “highest confidence origins”. Compared with shared origins or ORC-binding sites alone, these 1,300 origins are even more co-localized with gene promoters (**Fig. 4a**), TF binding hotspots (**Fig. 4b**) and constitutive CTCF binding sites (**Fig. 4c**). The GC content is not significantly higher for these “highest confidence origins” (**Fig. 4d**), and neither was their overlap with G-quadruplexes (**Fig. 4e**).

To understand the correlation between the different classes of origins and replication timing, we calculated the replication timing score for the origins following the ENCODE pipeline (Navarro Gonzalez et al. 2021; Hansen et al. 2010), and found that the mean replication timing score of shared origins suggest that they replicate earlier in S phase compared to the union origins, but there is not much difference between shared origins separated by whether they overlapped with ORC binding sites or not (**Fig. 4f**).

This suggests that even though shared origins near ORC binding sites are more enriched in TSS, TF binding hotspots and CTCF binding sites they are still similar to shared origins globally in their localization in early replicating, epigenetically active parts of the genome.

To investigate if the “highest confidence origins” (shared origins near ORC binding sites) are associated with higher transcriptional activity, we divided all human genes into four groups based on their expression level in the human K562 cell line and found the genes co-localized with the highest confidence origins exhibit higher expression levels compared to genes co-localized with all shared origins or union origins (**Fig. 4g**).

BART analysis (Wang et al. 2018) of which protein binding sites are most enriched in 743 origins overlapping with any ORC binding sites shows, as expected, that ORC2 binding sites are enriched near these origins. The other proteins bound near these origins are either transcriptional activators like EPC1, LEF1, ELK, EGR1, PAF1 or transcriptional repressors like KLF13, KRAB, ZNF639 or ZBTB33 (**Fig. 4h**). These results suggest that the shared origins overlapping with ORC binding sites tend to be more associated with transcriptional regulation than all shared origins.

Overlap of MCM binding sites with shared origins to define another type of highest confidence origins

The six subunit minichromosome maintenance complex (MCM) is loaded on chromatin in G1 and forms the core of the active helicase that unwinds the DNA to initiate DNA replication (Madine et al. 2000). Since MCM2-7 may be loaded by ORC and move away from ORC to initiate DNA replication, it could be expected that even if the ORC binding sites are not proximate to the 20,250 shared origins, they will be proximate with known MCM binding sites. To test this, we analyzed 18 human MCM ChIP datasets (ENCODE Project Consortium 2012; Ivanov et al. 2018; Utani et al. 2017). We identified 11,394 total MCM3-7 binding sites (union) and 3,209 shared MCM binding sites that are defined by an intersection of MCM3, MCM5 and MCM7 union peaks. Overall MCM binding sites displayed very similar genomic features as ORC binding sites (**Fig. 5a-e**). We then defined the genomic features common among the shared origins that are close to (≤ 1 kb) MCM binding sites. Like ORC-designated origins (**Fig. 4b**), the MCM-designated origins showed higher overlap with TF binding hotspots (**Fig. 5f**). Very interestingly, similar to the high overlap between yeast origins and yeast ORC sites (**Fig. 3f**), around 95% of yeast origins (Lee et al. 2021; Gros et al. 2015) are close to (≤ 1 kb) the union of all yeast MCM sites. In contrast, only ~4.5% shared human origins are close to (≤ 1 kb) the union of experimentally defined human MCM binding sites (**Fig. 5g**).

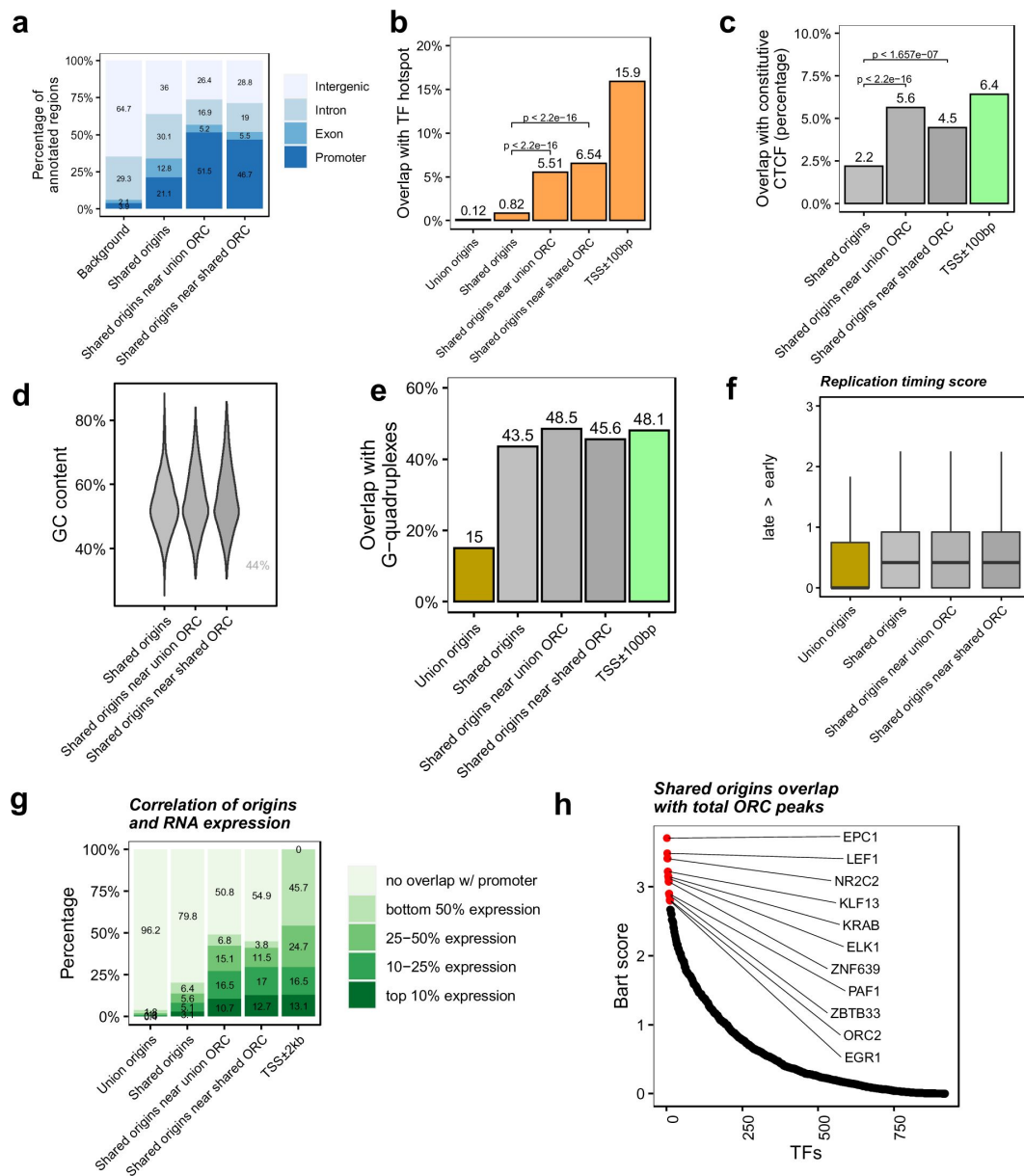


Figure 4

Shared ORC binding sites are more correlated with activate transcription.

(a) Genomic annotation of shared origins and shared origins that near (less than 1kb) ORC binding sites. **(b)** Overlap with TF hotspots of shared origins and shared origins that near ORC binding sites. **(c)** Overlap with constitutive CTCF binding sites of shared origins and shared origins that near ORC binding sites. **(d)** GC content of shared origins and shared origins that near ORC binding sites. **(e)** Overlap with G-quadruplex sites of shared origins and shared origins that near ORC binding sites. **(f)** Y-axis: Replication timing score (from Navarro, 2021) for indicated classes of origins. **(g)** Annotation of expression level of genes that overlapped with different groups of origins. **(h)** BART prediction of TFs associated with highest confidence origins.

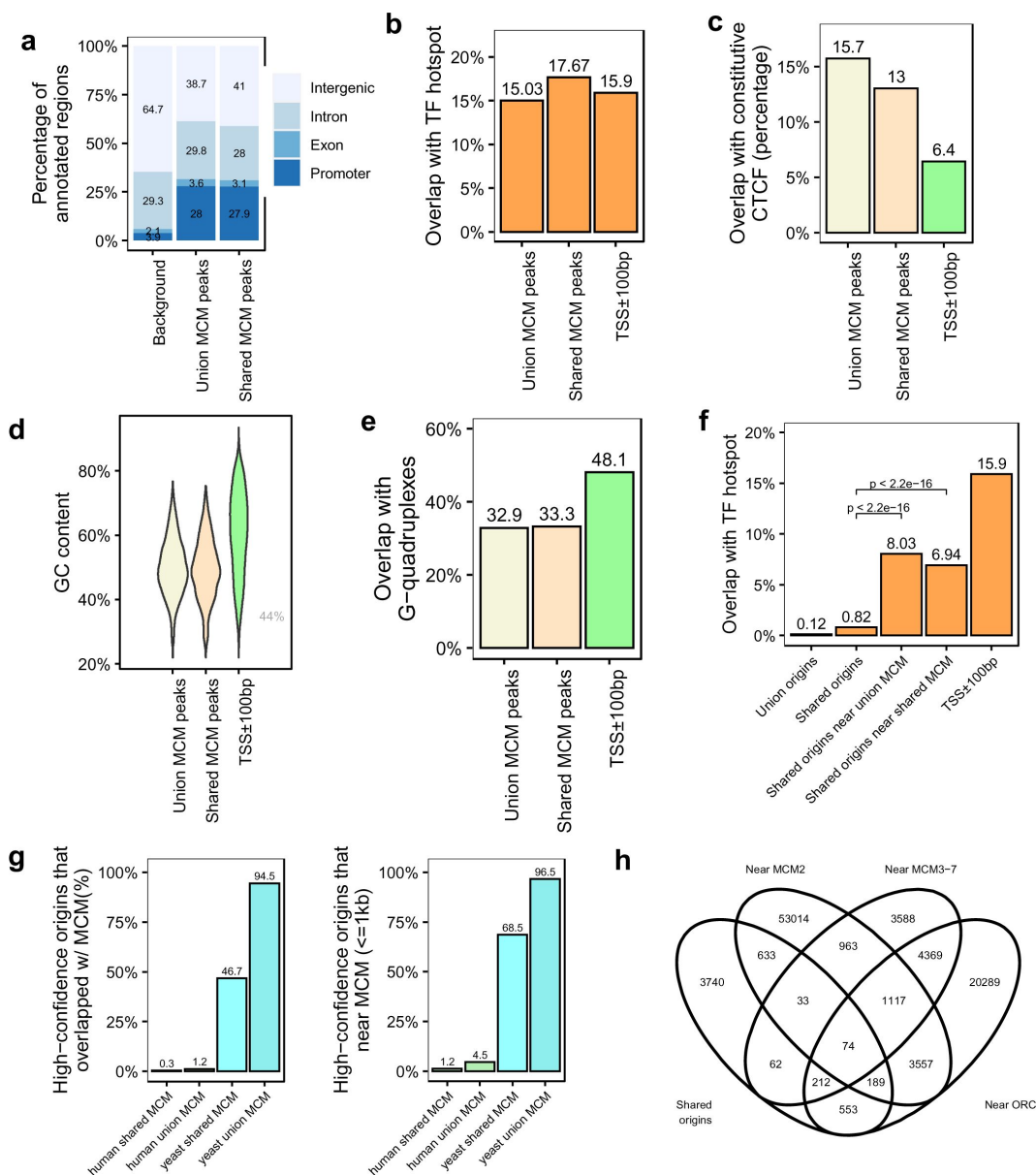


Figure 5

Genomic features of shared MCM binding sites.

(a) Genomic annotation of union MCM and shared MCM binding sites. (b) Overlap with TF hotspot of union MCM and shared MCM binding sites. (c) Overlap with constitutive CTCF binding rates of union MCM and shared MCM binding sites. (d) GC content of union MCM and shared MCM binding sites. (e) Overlap with G-quadruplex of union MCM and shared MCM binding sites. (f) Overlap with TF hotspots of shared origins and shared origins that near MCM binding sites. (g) The percentage of high-confidence origins (shared origins in human and confirmed origins in yeast) that overlapped with / near (region boundary less than 1kb) corresponding MCM binding sites. (h) Several confident origins defined by different methods. Number is calculated from total peak from each methods overlapping with total peak of all methods.

We examined all three types of high confidence origins (ORC designated, MCM3-7 designated and MCM2 designated) in a Venn Diagram (**Fig. 5h**) and identified 74 that were reproducibly identified by multiple methods (shared origins, near shared ORC binding sites, overlapping with MCM3-7 binding sites and MCM2 binding sites). The coordinates of these origins and their supporting data (ORC and MCM binding sites) are listed in Supplementary Table 5, and genome browser views for three of them are shown in **Fig. 6** to indicate the relationship between the coordinates.

Discussion

Through integrative analysis of publicly available 113 profiles of replication origins from multiple techniques, we identified ~7.5 million union origins in the human genome, of which only 20,250 shared origins are reproducibly identified by at least 20 SNS-seq datasets and confirmed by each of three other techniques. The following conclusions were reached: the shared (highly reproducible) origins are only a small subset of all origins identified and are in epigenetically active chromatin with a strong preference for promoters. However, G-quadruplexes, CTCF binding sites and ORC binding sites are also enriched near promoters, so that although these genomic features are enriched near the shared origins, it is not clear whether these genomic features actually specify origins, or whether they are simply enriched near origins because origins are near promoters. Finally, our results suggest that the co-localization of the shared origins with ORC or MCM2-7 binding sites is very low, much lower than in yeast, and so (a) the currently known binding sites of these proteins should not be used as surrogate markers of origins in human cells and (b) there is a great need to improve our methods of identifying isoforms of these proteins that are strictly functional for origin firing. The current models of origin firing require ORC to bind near origins, to load MCM2-7 nearby, and the latter to be converted to active helicase to fire origins. To rigorously test this model, we need examples of reproducible origins where these conditions are met, and our study helps identify 74 such sites where such experimental verification can be carried out.

Careful analysis shows that individual methods produce origin datasets that are more correlated with each other than across methods. The extensive variation of origins between datasets is likely because of background noise in all current techniques, and because there is extreme stochasticity of origin selection during DNA replication in individual cells and cell cycles in the same population of cells from a human cell line in culture. There are also differences that are created due to differences in cell lineage. Despite this, we could cut through those differences to focus on the small subset of shared origins that seem to be active in multiple cell lineages.

The shared origins are enriched with active histone marks and enriched in early replicating parts of the genome that are overwhelmingly in an active epigenetic environment. H3K4me3 has been reported to be enriched at replication origins (Picard et al. 2014; Cayrou et al. 2015), and it has also been reported that demethylation of H3K4me3 by KDM5C/JARID1C promotes origin active (Rondinelli et al. 2015). H3ac/H4ac are also reported to be enriched at replication origins (Cadoret et al. 2008; Sequeira-Mendes et al. 2009) and this is regulated by various histone acetyltransferases (HATs) and histone deacetylases (HDACS) (Goren et al. 2008; Wang et al. 2009). Interestingly, H3K27me3 has also been reported to be enriched at replication origins (Picard et al. 2014; Cayrou et al. 2015). The higher enrichment of *all* activating histone modifications relative to the repressive histone modifications (including H3K27me3) (**Fig. 2j**) rather than individual types of modification, suggest that the shared, highly reproducible origins, are preferentially in epigenetically active euchromatin, this aligns well with the general enrichment of shared origins with TSSs and TF hotspots, which are concentrated in gene-dense, epigenetically open parts of the chromosomes. This also aligns well with the fact that shared origins were not enriched for H4K20me3 and H3K9me3. H4K20me3 has been reported to be enriched near late-firing, heterochromatic initiation zones (Kirstein et al. 2021) and H3K9me3

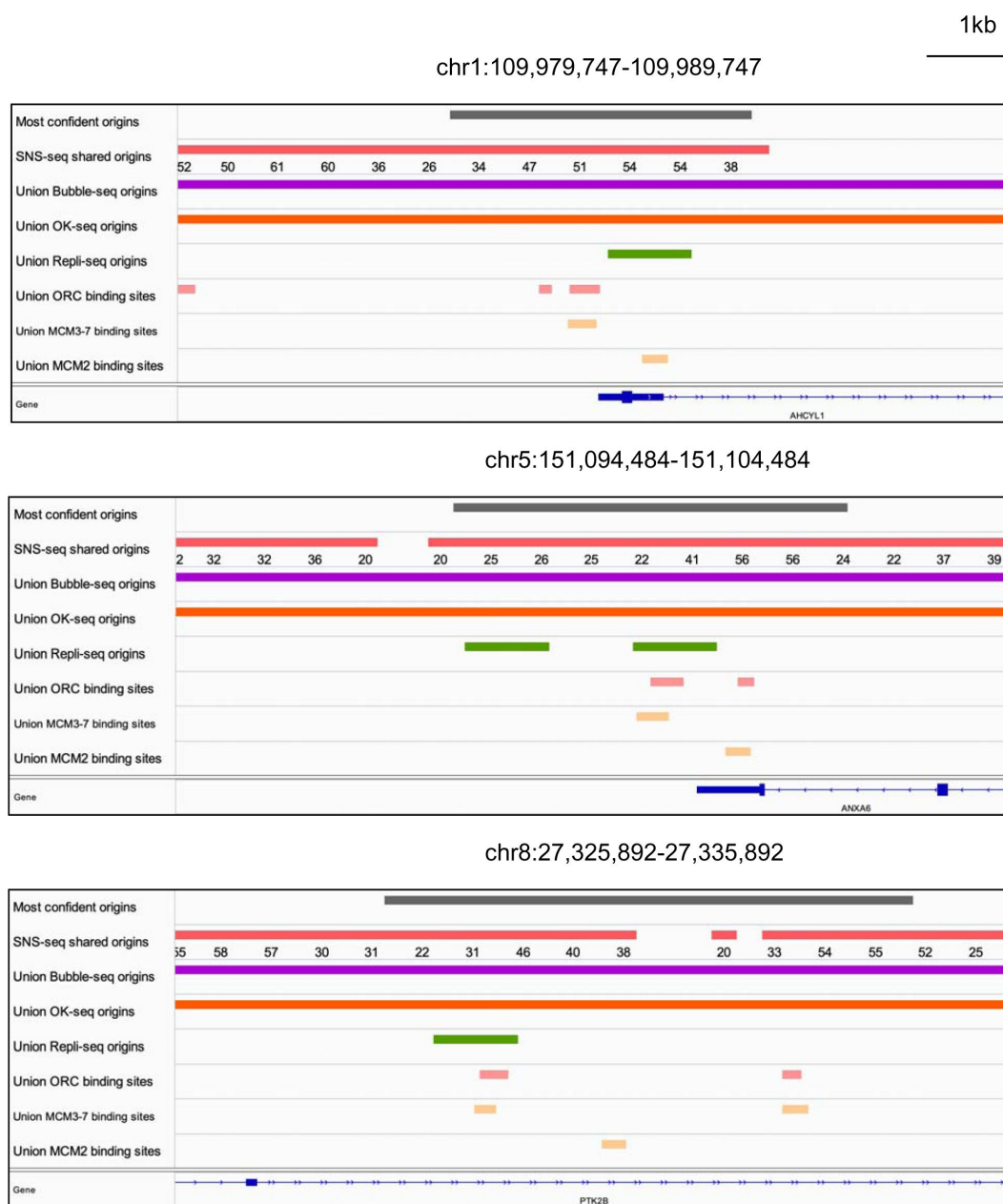


Figure 6

Genome browser screenshot for three of the 74 origins from Fig. 5h [↗](#)

The number on SNS-seq shared origins track is occupancy score of the origin.

enriched near late replicating origins (Wang et al. 2017 [↗](#)), while the shared origins are biased towards those that are in euchromatin. That origins, like Transcription start sites, prefer areas of the chromatin marked by activating epigenetic marks has been reviewed in 2016 (Prioleau and MacAlpine 2016 [↗](#)). We arrive at the same conclusion even though we worked with origins identified after this review, presumably with significant improvements in method and even though we focused on the most reproducible origins.

We characterized the genomic features of origins and found the shared origins are more co-localized with TSS. **Table 1** [↗](#) shows the summary of the overlap of union origins, shared origins, shared origins overlapping ORC binding sites with various genomic features and compares this with the overlap of TSS with the same genomic features. Overall, these results suggest that as we proceed from all origins, to shared origins to highest confidence origins that are proximate to ORC binding sites, we see increasing overlap with promoters and early replicating, transcriptionally active, epigenetically open parts of the genome and that features thought to be important for origin selection (like G-quadruplexes, CTCF binding sites, active chromatin, ORC binding sites) could simply co-occur with origins because of their enrichment near promoters (TSS).

To rigorously test whether the overlaps of shared origins that we observe with various genomic features are significantly above background, we performed a permutation test that evaluates whether the observed overlap is significantly above the mean expected overlap when the experimental dataset is randomized 1,000 times (**Table 2** [↗](#)). The overlap of *shared origins* with promoters (TSS±4Kb), G-quadruplexes, R-loops, and CTCF binding sites are all significantly above background. However, the overlap of *promoters* with G-quadruplexes, R loops, and CTCF binding sites are also all significantly above background, though the enrichment of G-quadruplexes is minimal. Thus the genomic features characterized are all significantly enriched in both origins and promoters, and may help to specify both, either independently or co-dependently.

G-quadruplexes are of particular interest because they have been experimentally shown to dictate origin-specification (Prorok et al. 2019 [↗](#); Valton et al. 2014 [↗](#)). Although there is a higher overlap of shared origins than union origins with G-quadruplexes (43.5% vs. 15%) this overlap is not higher than that between promoters and G-quadruplexes (48.1%) (**Fig. 2g** [↗](#)). This may suggest that the overlap between origins and G-quadruplexes may be secondary to the overlap between origins and promoters. However, the overlap of shared origins with G-quadruplexes is 3.5-fold above random while that of promoters with G quadruplexes is only 1.3-fold above random, which would be consistent with the idea that G-quadruplexes have a role in specifying origins and the overlap is not simply due to the proximity of origins with promoters.

Similarly, CTCF binding sites have been proposed to contribute to origin-specification (Emerson et al. 2022 [↗](#)). Here again, the 2.2% of shared origins that overlap with constitutive CTCF binding sites is less than the 6.4% of promoters that overlap with CTCF binding sites (**Fig. 2e** [↗](#)). Here, though, the permutation test in **Table 2** [↗](#) reveals that the enrichment of origins near CTCF binding sites (4.3-fold above random background) is comparable to the enrichment of promoters near the same sites (4.5-fold). This may, thus, suggest that the overlap of CTCF binding sites with origins is secondary to the overlap of origins with promoters.

Since ORC is an early factor for initiating DNA replication, shared human origins should be proximate to the reproducible ORC binding sites. The vast majority of ORC binding sites are not proximate to the shared origins and conversely, only about 6.4% of the shared origins are proximate (within 1 kb) to the reproducible (shared) ORC binding sites (**Fig. 3f** [↗](#)). Even with the most relaxed criteria nearly 85% of the most reproducible origins in human cells are not proximate to any ORC binding site (union, **Fig. 3f** [↗](#)). This low level of proximity contrasts with the ~100% of yeast origins that are within 1 kb of a yeast ORC binding site. Even when we examined cell line-specific origins with the ORC ChIP-seq datasets from the same cell lines, the co-localization between origins and ORC binding sites remained poor (Supplementary Fig. S5). Another study also

		% Overlap with				
	Number	Promoter	TF hotspot	CTCF	GC content	G quad
Union Origins	7,459,709	4.1	0.1	0.7	41.3333	15
Shared Origins	20,250	21.1	0.82	2.2	53	43.5
Shared Ori with shared ORC	1,300	46.7	6.54	4.5	54	45.6
TSS+/-100	26,237	100	15.9	6.4	61.5	48.1
Shared ORC	12,712	32.2	20.7	10	53.1	44.1

Table 1

Overlap of origins, TSS and ORC binding sites with indicated features.

Set_A	Set_B	#Set_A	#Set_B	Observed Overlap	# Random Iterations	Mean Random Overlap	Obs/Random Fold Enrichment	P-Value
origins	4k±promoters	20250	26237	4275	1000	649	6.6	< 0.001
origins	ORC_peaks	20250	12712	347	1000	54	6.4	< 0.001
origins	R-loop_zones	20250	59176	2607	1000	299	8.7	< 0.001
origins	G-quadruplex	20250	1444095	8818	1000	2529	3.5	< 0.001
origins	CTCF_sites	20250	22097	443	1000	104	4.3	<0.001
promoters	ORC_peaks	26237	12712	4832	1000	454	10.6	<0.001
promoters	R-loop_zones	26237	59176	10068	1000	1404	7.2	<0.001
promoters	G-quadruplex	26237	1444095	21809	1000	17242	1.3	<0.001
promoters	CTCF_sites	20250	22097	3644	1000	813	4.5	<0.001

Table 2

Permutation test of overlap of shared origins or promoters (TSS) with region around promoters, Shared ORC peaks (in > 2 datasets), R-loops, G-quadruplexes and CTCF binding sites. Fold enrichment of observed overlap relative to the mean overlap seen with 1000 randomizations of set A is indicated together with the p-value of the enrichment.

noted that only 13% of SNS-seq origins in K562 cells are near ORC2 binding sites (Miotto et al. 2016 [\[1\]](#)), but the authors suggested that this could occur if many SNS-seq sites are not real origins but arose from DNA breaks. A vast contamination of SNS-seq origins by sites of DNA breaks is discounted in our analysis because of the reproducibility of the shared origins in multiple labs, multiple lineages and multiple different techniques.

Instead of an unbiased determination of origins mapped by multiple groups to create a small set of reproducible origins, we could also empirically select a few well-curated origin datasets. Towards that end we used the Core origins identified by lambda exonuclease SNS-seq (Akerman et al. 2020 [\[2\]](#)) and the Ini-seq2 origins identified by labeling nuclei *in vitro* to identify the earliest replicated parts of the genome (Guilbaud et al. 2022 [\[3\]](#)). The shared origins were only a small subset of these origin datasets (Supplementary Fig. S8a and S8c). The percentage of these origins that were near ORC was still far less than what we observed in yeast: 13.7% for core origins and 30.4% for Ini-seq2 origins (Supplementary Fig. S8b and S8d). The higher overlap with Ini-seq2 origins could be because *in vitro* initiation of replication in isolated nuclei in Ini-seq preferentially identifies origins in early replicating, euchromatic parts of the genome, areas that have already been reported to be more enriched for ORC2 binding sites (Miotto et al. 2016 [\[1\]](#)).

There could be other explanations for the poor overlap of shared origins with ORC binding sites. We tested whether increasing the stringency of reproduction in SNS-seq data will produce more reproducible origins that are better co-localized with ORC binding sites. However, when we reduced the shared origins based on their reproduction by 30, 40, or 50 (out of 66) SNS-seq datasets, we did not see a marked improvement of their co-localization with currently known ORC binding sites (Supplementary Fig. S9a and S9b).

Finally, the permutation test in **Table 2** [\[4\]](#) demonstrates that the co-localization of origins with ORC binding sites was 6.5-fold ($p < 0.001$) greater than random expectation but this was less than the co-localization of promoters with ORC binding sites, 10.6-fold greater than random ($p < 0.001$). Thus, the co-localization data cannot conclusively say whether ORC binding near an origin is because ORC specifies origins, or whether this is secondary to origins being co-localized near promoters. Note, that the co-localization of ORC1 binding sites and origins in HeLa cells with highly active TSS has been noted in the past (Dellino et al. 2013 [\[5\]](#)). In human lymphoblast Raji cells ORC was enriched in promoters, and MCM depleted from gene bodies (Kirstein et al. 2021 [\[6\]](#)). ORC1 has also been reported to bind to RNA near promoters and stimulate replication from such sites (Mas et al. 2023 [\[7\]](#)). Thus our observation that ORC is highly enriched near promoters is expected, the main advance being that the random permutation test suggests that co-localization of TSS with ORC is more than the co-localization of shared origins with ORC. The question arises, are our results contradicting past studies. To the best of our knowledge, no one has taken such a comprehensive and quantitative approach to assess the proximity of origins with ORC binding sites genome-wide. We determined the percentage of origins that are within 1 kb of ORC-binding sites genome-wide and did a random permutation test to test whether the observed overlap is greater than random expectation. This told us that only a small percentage of the origins are within 1 kb of currently known ORC binding sites on a genome-wide scale, although selected browser shots as in **Fig. 6** [\[8\]](#) might suggest otherwise. Furthermore, even though Origins were enriched near ORC binding sites relative to random expectation this enrichment is not more than that of TSS near ORC binding sites.

Taken together, these results suggest three possibilities: (a) the ORC ChIP-seq or origin determination datasets in humans are very noisy and that the ChIP-seq data in particular fail to identify functional ORC binding sites; (b) that the MCM2-7 loaded at the ORC binding sites move very far to initiate origins, farther than 1 kb from the ORC binding sites; or (c) that there are as yet unexplored mechanisms by which many of the most reproducible origins are specified, mechanisms that might include ORC-independent modes of origin specification, similar to the

DnaA independent modes of origin specification in bacteria that use R-loops or DNA breaks to initiate replication (Itoh and Tomizawa 1980 [↗](#); Leela et al. 2021 [↗](#); Goswami and Gowrishankar 2022 [↗](#)).

There is significant evidence for possibility (b) above. MCM2-7 is either loaded far from ORC binding sites or move significant distances after loading in *Xenopus* egg extracts (Harvey and Newport 2003 [↗](#)). In yeasts there is evidence that MCM2-7 are pushed far away from the bodies of transcriptionally active genes by the RNA polymerase II and initiate replication at sites distant from where they are loaded on chromatin by ORC (Gros et al. 2015 [↗](#)). However, it is worth noting that despite this nearly 100% of the origins in yeast are within 1 kb of ORC binding sites (**Fig. 3f** [↗](#)). Similar observations have been made in *Drosophila*, where cyclin E/cdk2 kinase activity promotes the loading of vast excess of MCM2-7 on chromatin relative to ORC, and the MCM2-7 complexes move away from their loading sites due to the activity of the transcriptional apparatus (Powell et al. 2015 [↗](#)).

Since most models of replication initiation propose that a stably bound MCM2-7 complex is converted to an active CMG helicase at the time of origin firing, we hoped that even if ORC binding sites are not necessarily close to the shared origins, MCM2-7 binding sites will be more proximate to the shared origins in human cells. However, only 4.5% of shared origins are near any MCM2-7 binding site in human cells and again this is in contrast to the 96.5% of origins in yeast being near any MCM2-7 binding sites (**Fig. 5g** [↗](#)). Even if we focus on the limited data from the same cell line (K562 or HCT116) only 3.8% (union origins) or 5.5% shared origins overlap with union MCM3-7 binding sites in K562 cells and 19.5% of union origins overlap with any MCM2 binding sites in HCT116 cells (Supplementary Figure S5b, S5c and S5e). As with ORC, we also did the analysis with two selected origin datasets, the core origins (Akerman et al. 2020 [↗](#)) and the Ini-seq2 origins (Guilbaud et al. 2022 [↗](#)) and found that still a small percentage of these origins (6.6% and 10.7%, respectively) were near any MCM2-7 binding sites. Finally, a recent paper reported the binding sites of phosphorylated isoforms of MCM2 (Thakur et al. 2022 [↗](#)), and since phosphorylation of MCM2 is a prerequisite for MCM2-7 being activated as a helicase, we asked whether any of the phospho-MCM2 binding sites showed better co-localization with the shared origins (Supplementary Fig. S7). Phospho-S108 MCM2 showed the best result among the phospho-isoforms tested, with 22% of the shared origins being proximate to the Phospho-S108 MCM2 binding sites. Thus, phosphorylation on S108 of MCM2 may indeed mark MCM2-7 complexes that become active helicases which fire origins, but even then the co-localization was seen in a disappointingly low percentage of shared origins.

MCM2-7 ChIP-seq data is biased by a lot of noise as human MCM2-7 slides around after initial loading and due to contamination from the actively replicating CMG helicase that moves all over the genome (and pauses at many sites) during S phase, but this should increase the number of MCM2-7 binding sites and not decrease the percentage of origins that are near MCM2-7 sites. The better co-localization of shared origins with phospho-S108 MCM2 suggests that the ChIP methods can be improved to identify a small but critical pool of MCM2-7 that is engaged with the DNA stably in a way that permits origin specification. Alternatively, mapping of the sites bound to the active CMG helicase (MCM2-7, CDC45 and GINS), before the active helicase moves too far from the initiation site, may better enrich for sites that are destined to be origins.

A byproduct of our analysis of ORC ChIP-seq data, was the discovery that ORC1 and ORC2 are not strictly co-located as expected for the subunits of one ORC complex. Only 1,363 (21%) of the 6,501 ORC1 peaks overlapped with the 29,930 ORC2 peaks (Supplementary Fig. S6a), and the vast majority (68%) of ORC1 peaks are far away (≥ 2 kb) from the closest ORC2 peaks (Supplementary Fig. S6b). As a positive control, we checked other proteins that form well-established complexes like SMARCA4 and ARID1A in SWI/SNF complex, SMC1A and SMS63 in cohesin complex, EZH2 and SUZ12 in PRC2 complex, and found that they show a high proportion of overlap between their peaks as expected (Supplementary Fig. S6a). A possible explanation of the low overlapping rate

between ORC1 and ORC2 peaks could be that the datasets are from different cell types (Supplementary Fig. S6c). However, even in ChIP-seq peaks from different cell types, we found continued high overlap between SMC1A and SMC3 peaks as would be expected for complexes that do not have high inter-cell-line variability in binding sites. In contrast, there was a low overlap between EZH2 and SUZ12 peaks taken from different cell-lines suggesting that for other complexes there is significant inter-cell-line variability in binding sites (Supplementary Fig. S6d). Thus, the lack of overlap between the ORC1 and ORC2 association sites on chromatin could be explained by inter-cell-line variation of binding sites for ORC (like the PRC2 complex) or could be explained by the ORC subunits binding to chromatin independent of one another. We prefer the latter explanation because of our experimental result that a substantial fraction of ORC2, 3, 4, 5 and 6 bind to chromatin in the absence of ORC1, and ORC2, 3 and 6 bind to chromatin normally in the absence of ORC5 (Park and Asano 2008 [\[1\]](#); Shibata et al. 2016 [\[2\]](#); Okano-Uchida et al. 2018 [\[3\]](#); Shibata and Dutta 2020 [\[4\]](#)). We and others have also noted that human ORC2, 3, 4 and 5 form a stable subcomplex, with much looser association of the subcomplex with ORC1 and ORC6 (Dhar et al. 2001 [\[5\]](#)). Note that lineage-specific variation of where ORC may bind to the chromatin or the binding of ORC subcomplexes to chromatin may explain why so few of the ORC binding sites overlap with the shared origins but does not explain why only 6.4% of the shared origins (identified reproducibly in different cell lines) is proximate to any ORC binding sites.

Our analysis provides a characterization of origins in the human genome using multi-source data in the public domain. The sequencing-based profiles from different techniques have different biases in detecting DNA replication origins and show different genomic features. As one adds criteria to identify the most reproducible experimentally determined origins that are close to ORC binding sites, the overlap with TSS, TF-hotspots, constitutive CTCF binding sites and G-quadruplexes increases (Table 1 [\[6\]](#)), but in general the overlap with the latter three genomic features does not exceed that of the same features with TSS. Thus, although the overlaps are statistically significant, this correlation analysis cannot determine whether the same genomic features *independently* specify origins of replication and promoters of transcription. This is a question that should be asked experimentally and the 74 experimentally reproduced origin datasets with proximity to ORC and MCM2-7 ChIP-seq sites provides a starting list for such experimental queries. Finally, although our datasets for origins were large, as with all integrative data analyses we are limited by the quality of the data, which can be improved as experimental techniques for both origin identification and ORC or MCM2-7 binding site determination continue to improve.

Methods

Data processing

We collected all publicly available human sequencing datasets of replication origin profiling and ORC ChIP-seq from GEO database (Barrett et al. 2013 [\[7\]](#)). Data details of the collected datasets can be found in Supplementary Table 1. Raw sequence data in fastq format were downloaded and processed as follows: FastQC (Andrews S 2010 [\[8\]](#)) (v0.11.5) was used for quality control and sequence data were then mapped to human genome (hg38) using bowtie2 (Langdon 2015 [\[9\]](#)) (v2.2.9). Sam files were converted into bam files using samtools (Li et al. 2009 [\[10\]](#)) (v1.12) and only high-quality reads (q-score ≥ 30) were retained for subsequent analyses. Peak calling for ORC ChIP-seq data was performed with MACS2 callpeak function (`--nomodel --extsize 150 -g hs -B -SPMR -q 0.05 --keep-dup 1`) (Zhang et al. 2008 [\[11\]](#)) (v2.1.4). Samples with more than 1000 peaks were kept as high-quality samples. SNS-seq, Bubble-seq, and Rerep-seq peak calling was performed with SICER (Zang et al. 2009 [\[12\]](#)) with different parameters based on the different resolution of technique: SNS-seq (W200 G600), Bubble-seq (W5000 G15000), Rerep-seq (W200 G600). The OK-seq defined IZs is generated from previous published papers (Petryk et al. 2016 [\[13\]](#)), (Wu et al. 2018 [\[14\]](#)) and the coordinates information is provided by Drs. Chunlong Chen and Olivier Hyrien. Repli-seq defined

origins is generated from UW Encode group and is accessible in GSE34399. To obtain union peaks, peaks from each technique and from all techniques were merged using Bedtools (Quinlan and Hall 2010) merge (v2.29.2). Union origins that are longer than 300bp were cut into separate origins of a maximum length of 300bp. At the end we obtained 7,459,709 union origins.

To compare the locations of origins between samples, we performed a principal component analysis (PCA), where the matrix (to which PCA is applied) is of size 7.5M (number of union origins) x 113 (samples), with a “1” in the (*i*,*j*) position if the *i*th origin was detected in the *j*th dataset. We also performed pairwise pearson correlation of the origins in all the datasets to determine the reproducibility of the results from a given technique and across techniques. The distance metric for heatmap is generated in R using function cor (with parameters: use = “pairwise.complete.obs”, method = “pearson”).

Identification of shared origins

The origins detected by each sample are very different, which makes the identification of common origins difficult. SNS-seq origins have the highest resolution, and so we start with them. We identified 5,069,432 union SNS-seq origins using the same approach. We focus on the union SNS-seq origins that appear above a threshold number of SNS-seq origin datasets (high confidence SNS-seq origins) and then determine those that overlap with any initiation zone identified by each of the three different techniques to delineate a set of “shared origins”, that initiate replication all the time and can be detected by multiple approaches. Note that this approach does not lose the high resolution of the SNS-seq origins, it merely finds those SNS-seq origins that are most reproduced and detected by each of the other 3 techniques.

To identify the most commonly shared origins between SNS-seq samples, we use an occupancy score for each union SNS-seq origin to show how many samples identify that specific origin. A higher occupancy score means the origin is more commonly present in many samples (Supplementary Fig. S2a). High confidence SNS-seq origins are those that occur in a sufficient number of SNS-seq samples at an FDR of >0.1. We used an exponential model for background (green line) and plotted the distribution of occupancy scores (blue dotted curve) for origins from all SNS-seq samples (Fig. 2a). The exponential model can be expressed as

$$y = A * e^{(K*(x+m))}$$

where *x* is the occupancy score, and *y* is the expected number of origins. Origins that are reproduced by multiple samples have higher occupancy scores than the background distribution. An empirical FDR of 0.1 was used to determine the cutoff of occupancy score so that the number of observed origins with occupancy score greater than the cutoff should be 10 times more than expected in the background model (Fang et al. 2020). The high-confidence SNS-seq origins were thus those with occupancy score ≥20 in all 66 SNS-seq samples. Detailed parameters can be found in Supplementary Table 2. Origins mapping to ChrM are removed from SNS-seq shared origins to avoid the interference of mitochondria DNA.

Finally, the high-confidence SNS-seq origins defined by our model that overlap with any IZs from OK-seq, any IZs from Bubble-seq, and any origins defined by Repli-seq are called “shared origins” (Fig. 2b). A total of 20,250 shared origins were identified.

K562 shared origins were identified as K562 SNS-seq origins that overlapped with any of the union OK-seq and Repli-seq peaks. HeLa shared origins were identified as SNS-seq origins present in all 3 SNS-seq samples in HeLa and overlapped with any of the union OK-seq and Repli-seq peaks.

Origins, ORC and MCM ChIP-seq data in yeast

Yeast origins data were obtained from OriDB (Nieduszynski et al. 2007 [\[1\]](#)). 829 replication origins mapped to the yeast genome sacCer1 were converted to the sacCer3 genome using UCSC LiftOver (Hinrichs et al. 2006 [\[2\]](#)). 289 experimentally confirmed origins, considered as high-confidence origins, were used for co-localization analysis with ORC and MCM binding sites in yeast. ORC and MCM ChIP-seq datasets were collected from the public domain (Supplementary Table 1) and processed using the same procedure as used for human data with reference genome version sacCer3. The shared MCM binding sites were defined as the MCM peaks that occur in all samples.

Enrichment of histone marks at shared origins

Histone modification ChIP-seq peak files were downloaded from CistromeDB (Zheng et al. 2019 [\[3\]](#)). A total of 5,711 peak files, each with over 5,000 peaks, covering 29 distinct histone modifications in different human cell types were used to interrogate the association between histone marks and shared origins, using union origins as control. The enrichment analysis was applied for each peak file by comparing the peak number in each histone modification peak file overlapping with shared origins versus the overlapping with union origins. The Odds ratio obtained from two-tailed Fisher's exact test was used as the enrichment score for each peak file (Fig. 2j [\[4\]](#)). Odds ratio and pvalue are calculated using python package scipy.
`(stats.fisher_exact([[shared_ori_overlap_with_hm, shared_ori_not_overlap_with_hm],
[all_ori_overlap_with_hm - shared_ori_overlap_with_hm, all_ori_not_overlap_with_hm -
shared_ori_not_overlap_with_hm]]))`

Co-localization analysis

We use co-localization analysis to define shared origins and show the overlapping of CTCF peaks, TF hotspots, TSS regions, G quadruplex peaks, etc. with origins. The co-localization analysis was performed using Bedtools (Quinlan and Hall 2010 [\[5\]](#)) `intersect -u`. At least 1 base pair of intersection is required to be defined as overlapping.

Distance to the closest feature

According to the fact that ORC usually binds to origins but the loaded MCM2-7 can shift before firing an origin, we defined ORC that binds near the origins +/- 1kb as ORC near origins. Bedtools `closest -d -t "first"` was used to find the closest peak/region and distance for a given region. Origins with a binding site of ORC no further than 1kb are selected as origins near ORC binding site.

G4 binding sites

G4(G-quadruplex) predicted motif sites are obtained from a published G4 motif predictor: G4Hunter (Bedrat et al. 2016 [\[6\]](#)). The 1,444,095 G4 motif coordinates is provided by Laurent.

Genomic annotation of promoter, exon, intron, and intergenic region coverage

The coordinates of TSS, exon, intron are from UCSC hg38 version (Navarro Gonzalez et al. 2021 [\[7\]](#)). Promoter regions are defined as TSS +/- 1kb. Intergenic regions are other genome regions excluding promoter, exon, and intron regions. Genomic annotation for peaks is defined by overlapping with those 4 types of regions by at least 1 bp.

Replication timing score

We used publicly available Repli-seq data in K562 cell line for different cell phase from ENCODE project (Navarro Gonzalez et al. 2021 [\[8\]](#)) to measure the replication timing of origins (Supplementary table 1). We used a previously defined weighted average score (Hansen et

et al. 2010²⁴) to combine signal from the six cell phases with the following formula: $\text{score} = (0.917 \cdot G1b) + (0.750 \cdot S1) + (0.583 \cdot S2) + (0.417 \cdot S3) + (0.250 \cdot S4) + (0 \cdot G2)$. Higher values correspond to earlier replication.

Permutation Test

The R (version 4.1.3) package named *regioner* (version 1.24.0) was used to statistically evaluate the associations between region sets with minor modification. The following parameters were used for running *regioner*. Number of iterations: 10,000. Evaluation function: `numOverlaps`. Randomization function: `randomizeRegions`. `randomizeRegions`: hg38. `non.overlapping`: TRUE. `mc.set.seed`: FALSE

Acknowledgements

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Supplementary data

Supplementary Figures S1-S9: Figure legend on same page as figure.

Supplementary Table 1, collected public data. The original reference for each dataset can be found on the GEO page for each dataset.

1. Metadata of collected public Origin data.
2. Sample number of each cell type.
3. Metadata of collected public ORC ChIP-seq data.
4. Metadata of collected public MCM ChIP-seq data.
5. Metadata of collected public ORC and MCM ChIP-seq data in yeast.

Supplementary Table 2, parameters of model for SNS-seq Parameters of exponential model for SNS-seq origins.

Supplementary Table 3, union and shared ORC binding sites

1. Union ORC ChIP-seq peaks, with coordinates and occupancy scores.
2. Shared ORC binding sites (defined as union ORC ChIP-seq peaks with an occupancy score ≥ 2).

Supplementary Table 4, highest confidence origins

1. Coordinates of union ORC ChIP-seq binding sites with shared origins in 1kb region.
2. Coordinates of shared origins with shared ORC binding sites in 1kb region.
3. Coordinates of shared origins with union ORC binding sites in 1kb region.

Supplementary Table 5, 74 most confident origins

Coordinates of 74 origins that were reproducibly identified by multiple methods (shared origins, near shared ORC binding sites, overlapping with MCM3-7 binding sites and MCM2 binding sites).

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Editors

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Reviewer #1 (Public Review):

In the best genetically and biochemically understood model of eukaryotic DNA replication, the budding yeast, *Saccharomyces cerevisiae*, the genomic locations at which DNA replication initiates are determined by a specific sequence motif. These motifs, or ARS elements, are bound by the origin recognition complex (ORC). ORC is required for loading of the initially inactive MCM helicase during origin licensing in G1. In human cells, ORC does not have a specific sequence binding domain and origin specification is not specified by a defined motif. There have thus been great efforts over many years to try to understand the determinants of DNA replication initiation in human cells using a variety of approaches, which have gradually become more refined over time.

In this manuscript Tian et al. combine data from multiple previous studies using a range of techniques for identifying sites of replication initiation to identify conserved features of replication origins and to examine the relationship between origins and sites of ORC binding in the human genome. The authors identify a) conserved features of replication origins e.g. association with GC-rich sequences, open chromatin, promoters and CTCF binding sites. These associations have already been described in multiple earlier studies. They also examine the relationship of their determined origins and ORC binding sites and conclude that there is no relationship between sites of ORC binding and DNA replication initiation. While the conclusions concerning genomic features of origins are not novel, if true, a clear lack of colocalization of ORC and origins would be a striking finding. However, the majority of the datasets used do not report replication origins, but rather broad zones in which replication origins fire. Rather than refining the localisation of origins, the approach of combining diverse methods that monitor different objects related to DNA replication leads to a base dataset that is highly flawed and cannot support the conclusions that are drawn, as explained in more detail below.

Methods to determine sites at which DNA replication is initiated can be divided into two groups based on the genomic resolution at which they operate. Techniques such as bubble-seq, ok-seq can localise zones of replication initiation in the range ~50kb. Such zones may contain many replication origins. Conversely, techniques such as SNS-seq and ini-seq can localise replication origins down to less than 1kb. Indeed, the application of these different approaches has led to a degree of controversy in the field about whether human replication does indeed initiate at discrete sites (origins), or whether it initiates randomly in large zones with no recurrent sites being used. However, more recent work has shown that elements of both models are correct i.e. there are recurrent and efficient sites of replication initiation in the human genome, but these tend to be clustered and correspond to the demonstrated initiation zones (Guilbaud et al., 2022).

These different scales and methodologies are important when considering the approach of Tian et al. The premise that combining all available data from five techniques will increase accuracy and confidence in identifying the most important origins is flawed for two principal reasons. First, as noted above, of the different techniques combined in this manuscript, only SNS-seq can actually identify origins rather than initiation zones. It is the former that matters

when comparing sites of ORC binding with replication origin sites, if a conclusion is to be drawn that the two do not co-localise.

Second, the authors give equal weight to all datasets. Certainly, in the case of SNS-seq, this is not appropriate. The technique has evolved over the years and some earlier versions have significantly different technical designs that may impact the reliability and/or resolution of the results e.g. in Foulk et al. (Foulk et al., 2015), lambda exonuclease was added to single stranded DNA from a total genomic preparation rather than purified nascent strands), which may lead to significantly different digestion patterns (ie underdigestion). Curiously, the authors do not make the best use of the largest SNS-seq dataset (Akerman et al., 2020) by ignoring these authors separation of core and stochastic origins. By blending all data together any separation of signal and noise is lost. Further, I am surprised that the authors have chosen not to use data and analysis from a recent study that provides subsets of the most highly used and efficient origins in the human genome, at high resolution (Guilbaud et al., 2022).

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Guilbaud G, Murat P, Wilkes HS, Lerner LK, Sale JE, Krude T (2022) Determination of human DNA replication origin position and efficiency reveals principles of initiation zone organisation. *Nucleic Acids Res*, 50: 7436-7450

Update in response to authors' comments on the original review:

While the authors have clarified their approach to some aspects of their analysis, I believe they and I are just going to have to disagree about the methodology and conclusions of this work. I do not find the authors responses sufficiently compelling to change my mind about the significance of the study or veracity of the conclusions. In my opinion, the method for identification of strong origins is not robust and of insufficient resolution. In addition, the resolution and the overlap of the MCM Chip-seq datasets is poor. While the conclusion of the paper would indeed be striking and surprising if true, I am not at all persuaded that it is based on the presented data.

Reviewer #2 (Public Review):

Tian et al. performed a meta-analysis of 113 genome-wide origin profile datasets in humans to assess the reproducibility of experimental techniques and shared genomics features of origins. Techniques to map DNA replication sites have quickly evolved over the last decade, yet little is known about how these methods fare against each other (pros and cons), nor how consistent their maps are. The authors show that high-confidence origins recapitulate several known features of origins (e.g., correspondence with open chromatin, overlap with transcriptional promoters, CTCF binding sites). However, surprisingly, they find little overlap between ORC/MCM binding sites and origin locations.

Overall, this meta-analysis provides the field with a good assessment of the current state of experimental techniques and their reproducibility, but I am worried about: (a) whether we've learned any new biology from this analysis; (b) how binding sites and origin locations can be

so mismatched, in light of numerous studies that suggest otherwise; and (c) some methodological details described below.

-- I understand better the inclusion/exclusion logic for the samples. But I'm still not sure about the fragments. As the authors wrote, there is both noise and stochasticity; the former is not important but the latter is essential to include. How can these two be differentiated, and what may be the expected overlap as a function of different stochasticity rates?

-- Many of the major genomic features analyzed have already been found to be associated with origin sites. For example, the correspondence with TSS has been reported before:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6320713/>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6547456/>

-- Line 250: The most surprising finding is that there is little overlap between ORC/MCM binding sites and origin locations. The authors speculate that the overlap between ORC1 and ORC2 could be low because they come from different cell types. Equally concerning is the lack of overlap with MCM. If true, these are potentially major discoveries that butts heads with numerous other studies that have suggested otherwise.

The key missing dataset is ORC1 and ORC2 ChIP-seq from the same cell type. This shouldn't be too expensive to perform, and I hope someone performs this test soon. Without this, I remain on the fence about how much existing datasets are "junk" vs how much the prevailing hypothesis about replication needs to be revisited. Nonetheless, the authors do perform a nice analysis showing that existing techniques should be carefully used and interpreted.

Reviewer #3 (Public Review):

Summary: The authors present a thought-provoking and comprehensive re-analysis of previously published human cell genomics data that seeks to understand the relationship between the sites where the Origin Recognition Complex (ORC) binds chromatin, where the replicative helicase (Mcm2-7) is loaded, and where DNA replication actually begins (origins). The view that these should coincide is influenced by studies in yeast where ORC binds site-specifically to dedicated nucleosome-free origins where Mcm2-7 can be loaded and remains stably positioned for subsequent replication initiation. However, this is most certainly not the case in metazoans where it has already been reported that chromatin binding sites of ORC and Mcm2-7 do not necessarily overlap, nor do they always overlap with origins. This is likely due to Mcm2-7 possessing linear mobility on DNA (i.e., it can slide) such that other chromatin-contextualized processes can displace it from the site in which it was originally loaded. Additionally, Mcm2-7 is loaded in excess and thus only a fraction of Mcm2-7 would be predicted to coincide with replication start sites. This study reaches a very similar conclusion of these previous studies: they find a high degree of discordance between ORC, Mcm2-7, and origin positions in human cells.

Strengths: The strength of this work is its comprehensive and unbiased analysis of all relevant genomics datasets. To my knowledge, this is the first attempt to integrate these observations. It also is an important cautionary tale to not confuse replication factor binding sites with the genomic loci where replication actually begins, although this point is already widely appreciated in the field.

Weaknesses: The major weakness of this paper is the lack of novel biological insight and that the comprehensive approach taken failed to provide any additional mechanistic insight regarding how and why ORC, Mcm2-7, and origin sites are selected or why they may not coincide.

The following is the authors' response to the original reviews.

eLife assessment:

This study reports a meta-analysis of published data to address an issue that is topical and potentially useful for understanding how the sites of initiation of DNA replication are specified in human chromosomes. The work focuses on the role of the Origin Recognition Complex (ORC) and the Mini-Chromosome Maintenance (MCM2-7) complex in localizing origins of DNA replication in human cells. While some aspects of the paper are of interest, the analysis of published data is in parts inadequate to allow for the broad conclusion that, in contrast to multiple observations with other species, sites in the human genome for binding sites for ORC and MCM2-7 do not have extensive overlap with the location of origins of DNA replication.

Public Reviews:

Reviewer #1 (Public Review):

*In the best genetically and biochemically understood model of eukaryotic DNA replication, the budding yeast, *Saccharomyces cerevisiae*, the genomic locations at which DNA replication initiates are determined by a specific sequence motif. These motifs, or ARS elements, are bound by the origin recognition complex (ORC). ORC is required for loading of the initially inactive MCM helicase during origin licensing in G1. In human cells, ORC does not have a specific sequence binding domain and origin specification is not specified by a defined motif. There have thus been great efforts over many years to try to understand the determinants of DNA replication initiation in human cells using a variety of approaches, which have gradually become more refined over time.*

In this manuscript Tian et al. combine data from multiple previous studies using a range of techniques for identifying sites of replication initiation to identify conserved features of replication origins and to examine the relationship between origins and sites of ORC binding in the human genome. The authors identify a) conserved features of replication origins e.g. association with GC-rich sequences, open chromatin, promoters and CTCF binding sites. These associations have already been described in multiple earlier studies. They also examine the relationship of their determined origins and ORC binding sites and conclude that there is no relationship between sites of ORC binding and DNA replication initiation. While the conclusions concerning genomic features of origins are not novel, if true, a clear lack of colocalization of ORC and origins would be a striking finding.

Response: Thank you. That is where the novelty of the paper lies.

However, the majority of the datasets used do not report replication origins, but rather broad zones in which replication origins fire. Rather than refining the localisation of origins, the approach of combining diverse methods that monitor different objects related to DNA replication leads to a base dataset that is highly flawed and cannot support the conclusions that are drawn, as explained in more detail below.

Response: We are using the narrowly defined SNS-seq peaks as the gold standard origins and making sure to focus in on those that fall within the initiation zones defined by other methods. The objective is to make a list of the most reproducible origins. Unlike what the reviewer states, this actually refines the dataset to focus on the SNS origins that have also been reproduced by the other methods in multiple cell lines. We have changed the last box of Fig. 1A to make this clearer: Shared origins = reproducible SNS-seq origins that are contained

in initiation zones defined by Repli-seq, OK-seq and Bubble-seq. This and the Fig. 2B (as it is) will make our strategy clearer.

Methods to determine sites at which DNA replication is initiated can be divided into two groups based on the genomic resolution at which they operate. Techniques such as bubble-seq, ok-seq can localise zones of replication initiation in the range ~50kb. Such zones may contain many replication origins. Conversely, techniques such as SNS-seq and ini-seq can localise replication origins down to less than 1kb. Indeed, the application of these different approaches has led to a degree of controversy in the field about whether human replication does indeed initiate at discrete sites (origins), or whether it initiates randomly in large zones with no recurrent sites being used. However, more recent work has shown that elements of both models are correct i.e. there are recurrent and efficient sites of replication initiation in the human genome, but these tend to be clustered and correspond to the demonstrated initiation zones (Guilbaud et al., 2022).

These different scales and methodologies are important when considering the approach of Tian et al. The premise that combining all available data from five techniques will increase accuracy and confidence in identifying the most important origins is flawed for two principal reasons. First, as noted above, of the different techniques combined in this manuscript, only SNS-seq can actually identify origins rather than initiation zones. It is the former that matters when comparing sites of ORC binding with replication origin sites if a conclusion is to be drawn that the two do not co-localise.

Response: We agree. So the reviewer should agree that our method of finding SNS-seq peaks that fall within initiation zones actually refines the origins to find the most reproducible origins. We are not losing the spatial precision of the SNS-seq peaks.

Second, the authors give equal weight to all datasets. Certainly, in the case of SNS-seq, this is not appropriate. The technique has evolved over the years and some earlier versions have significantly different technical designs that may impact the reliability and/or resolution of the results e.g. in Foulk et al. (Foulk et al., 2015), lambda exonuclease was added to single stranded DNA from a total genomic preparation rather than purified nascent strands, which may lead to significantly different digestion patterns (ie underdigestion). Curiously, the authors do not make the best use of the largest SNS-seq dataset (Akerman et al., 2020) by ignoring these authors separation of core and stochastic origins. By blending all data together any separation of signal and noise is lost. Further, I am surprised that the authors have chosen not to use data and analysis from a recent study that provides subsets of the most highly used and efficient origins in the human genome, at high resolution (Guilbaud et al., 2022).

Response: 1) We are using the data from Akerman et al., 2020: Dataset GSE128477 in Supplemental Table 1. We have now separately examined the core origins defined by the authors to check its overlap with ORC binding (Supplementary Fig. S8b).

1. To take into account the refinement of the SNS-seq methods through the years, we actually included in our study only those SNS-seq studies after 2018, well after the lambda exonuclease method was introduced. Indeed, all 66 of SNS-seq datasets we used were obtained after the lambda exonuclease digestion step. To reiterate, we recognize that there may be many false positives in the individual origin mapping datasets. Our focus is on the True positives, the SNS-seq peaks that have some support from multiple SNS-seq studies AND fall within the initiation zones defined by the independent means of origin mapping (described in Fig. 1A and 2B). These True positives are most likely to be real and reproducible origins and should be expected to be near ORC binding sites.

We have changed the last box of Fig. 1A to make this clearer: Shared origins = reproducible SNS-seq origins that are contained in initiation zones defined by Repli-seq, OK-seq or Bubble-seq.

Ini-seq by Torsten Krude and co-workers (Guillbaud, 2022) does NOT use Lambda exonuclease digestion. So using Ini-seq defined origins is at odds with the suggestion above that we focus only on SNS-seq datasets that use Lambda exonuclease. However, Ini-seq identifies a much smaller subset of SNS-seq origins, so, as requested, we have also done the analysis with just that smaller set of origins, and it does show a better proximity to ORC binding sites, though even then the ORC proximate origins account for only 30% of the Ini-seq2 origins (Supplementary Fig. S8d). Note Ini-seq2 identifies DNA replication initiation sites seen in vitro on isolated nuclei.

References:

Akerman I, Kasaai B, Bazarova A, Sang PB, Peiffer I, Artufel M, Derelle R, Smith G, Rodriguez-Martinez M, Romano M, Kinet S, Tino P, Theillet C, Taylor N, Ballester B, Méchali M (2020) A predictable conserved DNA base composition signature defines human core DNA replication origins. *Nat Commun*, 11: 4826

Foulk MS, Urban JM, Casella C, Gerbi SA (2015) Characterizing and controlling intrinsic biases of lambda exonuclease in nascent strand sequencing reveals phasing between nucleosomes and G-quadruplex motifs around a subset of human replication origins. *Genome Res*, 25: 725-735

Guillbaud G, Murat P, Wilkes HS, Lerner LK, Sale JE, Krude T (2022) Determination of human DNA replication origin position and efficiency reveals principles of initiation zone organisation. *Nucleic Acids Res*, 50: 7436-7450

Reviewer #2 (Public Review):

Tian et al. perform a meta-analysis of 113 genome-wide origin profile datasets in humans to assess the reproducibility of experimental techniques and shared genomics features of origins. Techniques to map DNA replication sites have quickly evolved over the last decade, yet little is known about how these methods fare against each other (pros and cons), nor how consistent their maps are. The authors show that high-confidence origins recapitulate several known features of origins (e.g., correspondence with open chromatin, overlap with transcriptional promoters, CTCF binding sites). However, surprisingly, they find little overlap between ORC/MCM binding sites and origin locations.

Overall, this meta-analysis provides the field with a good assessment of the current state of experimental techniques and their reproducibility, but I am worried about: (a) whether we've learned any new biology from this analysis; (b) how binding sites and origin locations can be so mismatched, in light of numerous studies that suggest otherwise; and (c) some methodological details described below.

Major comments:

- *Line 26: "0.27% were reproducibly detected by four techniques" -- what does this mean? Does the fragment need to be detected by ALL FOUR techniques to be deemed reproducible?*

Response: If the reproducible SNS-seq peaks are included in the reproducible initiation zones found by the other methods, then we consider it reproducible across datasets. The strategy is

to focus our analysis on the most reproducible SNS-seq peaks that happen to be in reproducible initiation zones. It is the best way to confidently identify a very small set of true positive origins. We have re-stated this in the abstract: “only 0.27% were reproducibly obtained in at least 20 independent SNS-seq datasets and contained in initiation zones identified by each of three other techniques (20,250 shared origins),...”

And what if the technique detected the fragment is only 1 of N experiments conducted; does that count as "detected"?

Response: A reproducible SNS-seq origin has been reproduced above a statistical threshold of 20 reproductions of SNS-seq datasets. A threshold of reproduction in 20 datasets out of 66 SNS-seq datasets gives an FDR of <0.1 . This is explained in Fig. 2a and Supplementary Fig. S2. For the initiation zones, we considered a Zone even if it appears in only 1 of N experiments, because N is usually small. This relaxed method for selecting the initiation zones gives the best chance of finding SNS-seq peaks that are reproduced by the other methods.

Later in Methods, the authors (line 512) say, "shared origins ... occur in sufficient number of samples" but what does sufficient mean?

Response: “Sufficient” means that SNS-seq origin was reproducibly detected in ≥ 20 datasets and was included in any initiation zone defined by three other techniques.

Then on line 522, they use a threshold of "20" samples, which seems arbitrary to me. How are these parameters set, and how robust are the conclusions to these settings? An alternative to setting these (arbitrary) thresholds and discretizing the data is to analyze the data continuously; i.e., associate with each fragment a continuous confidence score.

Response: We explained Fig. 2a and Supplementary Fig. S2 on line 192 as follows: The occupancy score of each origin defined by SNS-seq (Supplementary Fig. 2a) counts the frequency at which a given origin is detected in the datasets under consideration. For the random background, we assumed that the number of origins confirmed by increasing occupancy scores decreases exponentially (see Methods and Supplementary Table 2). Plotting the number of origins with various occupancy scores when all SNS-seq datasets published after 2018 are considered together (the union origins) shows that the experimental curve deviates from the random background at a given occupancy score (Fig. 2a). The threshold occupancy score of 20 is the point where the observed number of origins deviates from the expected background number (with an FDR < 0.1) (Fig. 2a).

In the Methods: We have revised the section, “Identification of shared origins” to better describe our strategy. The number of observed origins with occupancy score greater than 20 (out of 66 measures) is 10 times more than expected from the background model. This approach is statistically sound and described by us in (Fang et al. 2020).

- *Line 20: "50,000 origins" vs "7.5M 300bp chromosomal fragments" -- how do these two numbers relate? How many 300bp fragments would be expected given that there are ~50,000 origins? (i.e., how many fragments are there per origin, on average)? This is an important number to report because it gives some sense of how many of these fragments are likely nonsense/noise. The authors might consider eliminating those fragments significantly above the expected number, since their inclusion may muddle biological interpretation.*

Response: We confused the reviewer by the way we wrote the abstract. The 50,000 origins that are mentioned in the abstract is the hypothetical expected number of origins that have to fire to replicate the whole 6×10^9 nt diploid genome based on the average inter-origin

distance of 100 kb (as determined by molecular combing). The 7.5M 300 bp fragments are the genomic regions where the 7.5M union SNS-seq-defined origins are located. Clearly, that is a lot of noise, some because of technical noise and some due to the fact that origins fire stochastically. Which is why our paper focuses on a smaller number of reproducible origins, the 20,250 shared origins. Our analysis is on the 20,250 shared origins, and not on all 7.5M union origins. Thus, we are not including the excess of non-reproducible (stochastic?) origins in our analysis.

The revised abstract in the revised paper will say: “Based on experimentally determined average inter-origin distances of ~100 kb, DNA replication initiates from ~50,000 origins on human chromosomes in each cell-cycle. The origins are believed to be specified by binding of factors like the Origin Recognition Complex (ORC) or CTCF or other features like G-quadruplexes. We have performed an integrative analysis of 113 genome-wide human origin profiles (from five different techniques) and 5 ORC-binding site datasets to critically evaluate whether the most reproducible origins are specified by these features. Out of ~7.5 million union origins identified by all the SNS-seq datasets, only 0.27% were reproducibly obtained in at least 20 independent SNS-seq datasets and contained in initiation zones identified by any of three other techniques (20,250 shared origins), suggesting extensive variability in origin usage and identification in different circumstances.”

- *Line 143: I'm not terribly convinced by the PCA clustering analysis, since the variance explained by the first 2 PCs is only ~25%. A more robust analysis of whether origins cluster by cell type, year etc is to simply compute the distribution of pairwise correlations of origin profiles within the same group (cell type, year) vs the correlation distribution between groups. Relatedly, the authors should explain what an "origin profile" is (line 141). Is the matrix (to which PCA is applied) of size 7.5M x 113, with a "1" in the (i,j) position if the ith fragment was detected in the jth dataset?*

Response: The reviewer is correct about how we did the PCA and have now included the description in the Methods. We have now done the pairwise correlations the way the reviewer suggests, and it is clear that each technique correlates best with itself (though there are some datasets that do not correlate as well as the others even with the same technique) (Supp. Fig. S3). We have also done the PCA by techniques (Fig. 1c), by cell types for all techniques (Supp. Fig. S1c), by cell-types for SNS-seq only (Supp. Fig. S1d), and by year of publication of SNS-seq data (Supp. Fig. S1e). Our conclusions remain the same: in general, origins defined from the same cell lineage are more similar to each other than across lineages, though this similarity within a lineage is more pronounced when we focus on SNS-seq alone. However, even when we look at SNS-seq alone, there is not a perfect overlap of origins determined by different studies on the same lineage. Finally, although we looked only at SNS-seq data after 2018, by which time lamda exonuclease had become the accepted way of defining SNS-seq, there is surprising clustering around each year.

- *It's not clear to me what new biology (genomic features) has been learned from this meta-analysis. All the major genomic features analyzed have already been found to be associated with origin sites. For example, the correspondence with TSS has been reported before:*

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6320713/>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6547456/>

So what new biology has been discovered from this meta-analysis?

Response: The new biology can be summarized as: (a) We can identify a set of reproducible (in multiple datasets and in multiple cell lines) SNS-seq origins that also fall within initiation zones identified by completely independent methods. These may be the best origins to study in the midst of the noise created by stochastic origin firing. (b) The overlap of these Shared origins (True Positive Origins) with known ORC binding sites is tenuous. So either all the origin mapping data, or all the ORC binding data has to be discarded, or this is the new biological reality in mammalian cancer cells: on a genome-wide scale the most reproduced origins are not in close proximity to ORC binding sites, in contrast to the situation in yeast. (c) Several of the features reported to define origins (CTCF binding sites, G quadruplexes etc.) could simply be from the fact that those features also define transcription start sites (TSS), and the origins may prefer to locate to these parts of the genome because of the favorable chromatin state, instead of the sequence or the structural features of CTCF binding sites or G quadruplexes specifically locating the origins.

- *Line 250: The most surprising finding is that there is little overlap between ORC/MCM binding sites and origin locations. The authors speculate that the overlap between ORC1 and ORC2 could be low because they come from different cell types. Equally concerning is the lack of overlap with MCM. If true, these are potentially major discoveries that butts heads with numerous other studies that have suggested otherwise. More needs to be done to convince the reader that such a mis-match is true. Some ideas are below:*

Idea 1) One explanation given is that the ORC1 and ORC2 data come from different cell types. But there must be a dataset where both are mapped in the same cell type. Can the authors check the overlap here? In Fig S4A, I would expect the circles to not only strongly overlap but to also be of roughly the same size, since both ORC's are required in the complex. So something seems off here.

Response: We agree with the reviewer that there is something “off here”. Either the techniques that report these sites are all wrong, or the biology does not fit into the prevailing hypothesis. As shown in Supplementary Fig. S6C, we do not have ORC1 and ORC2 ChIP-seq data from the same cell-type. We have ORC1 ChIP-seq and SNS-seq data from HeLa cells and ORC2 ChIP seq and origins from K562 cells, and so have now done the overlap of the binding sites to the shared origins in the same cell-type in the new Figure S5e and S5f. Out of 9605 shared origins in K562 cells, 12.8% overlap with ORC2 and 5.4% overlap with MCM3-7 binding sites also defined in K562 cells. Out of 8305 shared origins in HeLa cells, 4.4% overlap with ORC1 binding sites defined in HeLa cells.

There is nothing in the Literature that shows that various ORC subunits ChIP-seq to the same sites, and we have unpublished data that shows very poor overlap in the ChIP binding sites of different ORC subunits. The poor overlap between the binding sites of subunits of the same complex either suggests that the subunits do not always bind to the chromatin as a six-subunit complex or that all the ORC subunit ChIP-seq data in the Literature is suspect. We provide in the supplementary figure S6A examples of true positive complexes (SMARCA4/ARID1A, SMC1A/SMC3, EZH2/SUZ12), whose subunits ChIP-seq to a large fraction of common sites.

Idea 2) Another explanation given is that origins fire stochastically. One way to quantify the role of stochasticity is to quantify the overlap of origin locations performed by the same lab, in the same year, in the same experiment, in the same cell type -- i.e., across replicates -- and then compute the overlap of mapped origins. This would quantify how much mis-match is truly due to stochasticity, and how much may be due to other factors.

Response: A given lab may have superior reproducibility with its own results compared to the entire field, and the finding that origins published in the same year tend to be clustered together could be because a given lab publishes a number of origin sets in a single paper in a given year. But the notion of stochasticity is well accepted in the field because of this observation: the average inter-origin distance measured by single molecule techniques like molecular combing is ~100 kb, but the average inter-origin distance measure on a population of cells (same cell line) is ~30 kb. The only explanation is that in a population of cells many origins can fire, but in a given cell on a given allele, only one-third of those possible origins fire. This is why we did not worry about the lack of reproducibility between cell-lines, labs etc, but instead focused on those SNS-seq origins that are reproducible over multiple techniques and cell lines.

Idea 3) A third explanation is that MCMs are loaded further from origin sites in human than in yeast. Is there any evidence of this? How far away does the evidence suggest, and what if this distance is used to define proximity?

Response: MCMs, of course, have to be loaded at an origin at the time the origin fires because MCMs provide the core of the helicase that starts unwinding the DNA at the origin. Thus, the lack of proximity of MCM binding sites with origins can be because the most detected MCM sites (where MCM spends the most time in a cell-population) does not correspond to where it is first active to initiate origin firing. This has been discussed. MCMs may be loaded far from origin site, but because of their ability to move along the chromatin, they have to move to the origin-site at some point to fire the origin.

Idea 4) How many individual datasets (i.e., those collected and published together) also demonstrate the feature that ORC/MCM binding locations do not correlate with origins? If there are few, then indeed, the integrative analysis performed here is consistent. But if there are many, then why would individual datasets reveal one thing, but integrative analysis reveal something else?

Response: In the revised manuscript we have now discussed Dellino, 2013; Kirstein, 2021; Wang, 2017; Mas, 2023. None of them have addressed what we are addressing, which is whether the small subset of the most reproducible origins proximal to ORC or MCM binding sites, but the discussion is essential.

Idea 5) What if you were much more restrictive when defining "high-confidence" origins / binding sites. Does the overlap between origins and binding sites go up with increasing restriction?

Response: We have made SNS-seq origins more restrictive by selecting those reproduced by 30, 40, or 50 datasets, in addition to the FDR-determined cutoff of 20. The number of origins fall, but when we do not see any significant increase in the % of origins that overlap with or are proximal to with all ORC or MCM binding sites or Shared ORC or MCM binding sites. This analysis is now included in Supp. Fig. S9 and discussed.

Overall, I have the sense that these experimental techniques may be producing a lot of junk. If true, this would be useful for the field to know! But if not, and there are indeed "unexplored mechanisms of origin specification" that would be exciting. But I'm not convinced yet.

- *It would be nice in the Discussion for the authors to comment about the trade-offs of different techniques; what are their pros and cons, which should be used*

when, which should be avoided altogether, and why? This would be a valuable prescription for the field.

Response: Thanks for the suggestion. We have done what the reviewer suggested in the new Supp. Fig. S4.

Among the 20,250 high-confidence shared origins, 9,901 (48.9%) overlapped with SNS-seq origins in K562; 3,872 (19.1%) overlapped with OK-seq IZs; 1,163 (5.7%) overlapped with Repli-seq IZs.

In the reciprocal direction, we asked which method best picks out the highly reproducible shared origins. 2.7% of SNS-seq origins, 17.2% of OK-seq initiation zones and 7.7% of Repli-seq initiation zones overlapped with the 20,250 shared origins

Thus SNS-seq identifies more of the reproducible origins, but it comes with a high false positive rate.

ORC ChIP-seq and MCM ChIP-seq data do not define origins: they define the binding sites of these proteins. Thus we have discussed why the ChIP-seq sites of these protein complexes should not be used to define origins.

Reviewer #3 (Public Review):

Summary: The authors present a thought-provoking and comprehensive re-analysis of previously published human cell genomics data that seeks to understand the relationship between the sites where the Origin Recognition Complex (ORC) binds chromatin, where the replicative helicase (Mcm2-7) is situated on chromatin, and where DNA replication actually begins (origins). The view that these should coincide is influenced by studies in yeast where ORC binds site-specifically to dedicated nucleosome-free origins where Mcm2-7 can be loaded and remains stably positioned for subsequent replication initiation. However, this is most certainly not the case in metazoans where it has already been reported that chromatin binding sites of ORC, Mcm2-7, and origins do not necessarily overlap, likely because ORC loads the helicase in transcriptionally active regions of the genome and, since Mcm2-7 retains linear mobility (i.e., it can slide), it is displaced from its original position by other chromatin-contextualized processes (for example, see Gros et al., 2015 Mol Cell, Powell et al., 2015 EMBO J, Miotto et al., 2016 PNAS, and Prioleau et al., 2016 G&D amongst others). This study reaches a very similar conclusion: in short, they find a high degree of discordance between ORC, Mcm2-7, and origin positions in human cells.

Strengths: The strength of this work is its comprehensive and unbiased analysis of all relevant genomics datasets. To my knowledge, this is the first attempt to integrate these observations and the analyses employed were suited for the questions under consideration.

Response: Thank you for recognizing the comprehensive and unbiased nature of our analysis. The fact that the major weakness is that the comprehensive view fails to move the field forward, is actually a strength. It should be viewed in the light that we cannot find evidence to support the primary hypothesis: that the most reproducible origins must be near ORC and MCM binding sites. This finding will prevent the unwise adoption of ORC or MCM binding sites as surrogate markers of origins and will stimulate the field to try and improve methods of identifying ORC or MCM binding until the binding sites are found to be proximal to the most reproducible origins. The last possibility is that there are ORC- or MCM-independent modes of defining origins, but we have no evidence of that.

Weaknesses: The major weakness of this paper is that this comprehensive view failed to move the field forward from what was already known. Further, a substantial body of relevant prior genomics literature on the subject was neither cited nor discussed. This omission is important given that this group reaches very similar conclusions as studies published a number of years ago. Further, their study seems to present a unique opportunity to evaluate and shape our confidence in the different genomics techniques compared in this study. This, however, was also not discussed.

Response: We have done what the reviewer suggested: use K562 cell type-specific data where origins have been defined by three methods and reporting the percent of shared origins identified by each method (Supp. Fig. S4). Thanks for the suggestion. We have discussed now that SNS-seq identifies more of the reproducible origins, but it comes with a high false positive rate. ORC ChIP-seq and MCM ChIP-seq data do not define origins: they define the binding sites of these proteins. Thus, we have discussed that the ChIP-seq sites of these protein complexes as we now have them should not be used to define origins.

We do not cite the SNS-seq data before 2018 because of the concerns discussed above about the earlier techniques needing improvement. We have discussed other genomics data that we failed to discuss.

We have cited the papers the reviewer names:

Gros, Mol Cell 2015 and Powell, EMBO J. 2015 discuss the movement of MCM2-7 away from ORC in yeast and flies and will be cited. MCM2-7 binding to sites away from ORC and being loaded in vast excess of ORC was reported earlier on *Xenopus* chromatin in PMC193934, and will also be cited.

Miotto, PNAS, 2016: publishes ORC2 ChIP-seq sites in HeLa (data we have used in our analysis), but do not measure ORC1 ChIP-seq sites. They say: “ORC1 and ORC2 recognize similar chromatin states and hence are likely to have similar binding profiles.” This is a conclusion based on the fact that the ChIP-seq sites in the two studies are in areas with open chromatin, it is not a direct comparison of binding sites of the two proteins.

Prioleau, G&D, 2016: This is a review that compared different techniques of origin identification but has no primary data to say that ORC and MCM binding sites overlap with the most reproducible origins. It has now been referenced in the context of epigenetic marks and origins.

Reviewing Editor:

While there is some disagreement between the reviewers about the analysis performed, there are relevant concerns about the data analyzed (reviewers 1 and 2) and the biological significance of the observation (all three reviewers). There is also concern raised about the ORC ChIP-Seq data and the lack of overlap between published data for ORC1 and ORC2, which, if they were in a complex, the overlap in binding sites should be much better than reported.

Given the high overlap of ChIP-seq data for subunits of three other complexes shown in Supp. Fig. S6A, the most likely explanation is that ORC1 and ORC2 do not necessarily bind to DNA only as part of a complex. In other words, other protein complexes that contain one subunit or the other also bind DNA. This is not entirely unexpected. Biochemically the ORC2-3-4-5 complex is more stable and more abundant than the six subunit ORC.

Reviewer #2 (Recommendations For The Authors):

Minor comments:

- Line 44, missing spaces near references: "origins(Hu". Repeated issue throughout the manuscript.
- Line 82: "Notably any technical biases are uniquely associated with each assay" -- how do you know the biases are unique to each assay and orthogonal to each other?
- Line 135: typo: "using pipeline"
- Line 136: "All the 113 datasets" -> "Each of the 113 datasets"?
- Line 156: "differences among different techniques" -> "different" can be removed.
- Figure 4F: I don't see any difference in 4F amongst shared *. What is the y-axis anyways?

We have addressed these issues in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

The most significant omission is a contextualization of the results in the discussion and an explanation of why these results matter for the biology of replication, disease, and/or our confidence in the genomic techniques reported on in this study. As written, the discussion simply restates the results without any interpretation towards novel insight. I suggest that the authors revise their discussion to fill this important gap.

A second important, unresolved point is whether replication origins identified by the various methods differ due to technical reasons or because different cell types were analyzed. Given the correlation between TSS and origins (reported in this study but many others too), it is somewhat expected that origins will differ between cell types as each will have a distinct transcriptional program. This critique is partly addressed in Figure S1C. However, given the conclusion that the techniques are only rarely in agreement (only 0.27% origins reproducibly detected by the four techniques), a more in-depth analysis of cell type specific data is warranted. Specifically, I would suggest that cell type-specific data be reported wherever origins have been defined by at least two methods in the same cell type, specifically reporting the percent of shared origins amongst the datasets. This type of analysis may also inform on whether one or more techniques produces the highest (or lowest) quality list of true origins.

We have done what has been suggested: used K562 cell type-specific data because here the origins have been defined by at least two methods in the same cell type, and reported the percent of shared origins amongst the datasets (Supp. Fig. S4).

Other MINOR comments include:

- Line 215: the authors show that shared origins overlap with TF binding hotspots more often than union origins, which they claim suggests "that they are more likely to interact with transcription factors." As written, it sounds like the authors

are proposing that ORC may have some direct physical interaction with transcription factors. Is this intended? If so, what support is there for this claim?

The reviewer is correct. We have rephrased because we have no experimental support for this claim.

- *In the text, Figure 3G is discussed before Figure 3F. I suggest switching the order of these panels in Figure 3.*

Done.

- *It's not clear what Figure 5H to Figure 6 accomplishes. What specifically is added to the story by including these data? Is there something unique about the high confidence origins? If there is nothing noteworthy, I would suggest removing these data.*

We want to keep them to highlight the small number of origins that meet the hypothesis that ORC and MCM must bind at or near reproducible origins. These would be the origins that the field can focus in on for testing the hypothesis rigorously. They also show the danger of evaluating proximity between ORC or MCM binding sites with origins based on a few browser shots. If we only showed this figure we could conclude that ORC and MCM binding sites are very close to reproducible origins.

- *Line 394: "Since ORC is an early factor for initiating DNA replication, we expected that shared human origins will be proximate to the reproducible ORC binding sites." This is only expected if one disbelieves the prior literature that shows that ORC and origins are not, in many cases, proximal. This statement should be revised, or the previous literature should be cited, and an explanation provided about why this prior work may have missed the mark.*

We do not know of any genome-wide study in mammalian cell lines where ORC binding sites and MCM binding have been compared to highly reproducible origins, or that show that these binding sites and highly reproducible origins are mostly not proximal to each other. Most studies cherry pick a few origins and show by ChIP-PCR that ORC and/or MCM bind near those sites. Alternatively, studies sometimes show a selected browser shot, without a quantitative measure of the overlap genome wide and without doing a permutation test to determine if the observed overlap or proximity is higher than what would be expected at random with similar numbers of sites of similar lengths. In the revised manuscript we have discussed Dellino, 2013; Kirstein, 2021; Wang, 2017; Mas, 2023. None of them have addressed what we are addressing, is the small subset of the most reproducible origins proximal to ORC or MCM binding sites?

- *Line 402-404: given the lack of agreement between ORC binding sites and origins the authors suggest as an explanation that "MCM2-7 loaded at the ORC binding sites move much further away to initiate origins far from the ORC binding sites, or that there are as yet unexplored mechanisms of origin specification in human cancer cells". The first part of this statement has been shown to be true (Mcm2-7 movement) and should be cited. But what do the authors mean by the second suggestion of "unexplored mechanisms"? Please expand.*

We have addressed this point in the revised manuscript.

- *The authors should better reference and discuss the previous literature that relates to their work, some of these include Gros et al., 2015 Mol Cell, Powell et al., 2015 EMBOJ, Miotto et al., 2016 PNAS, but likely there are many others.*

We have addressed this point in the revised manuscript.